INCH-POUND

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MILITARY SPECIFICATION

TRUCKS, LIFT, FORK, ELECTRIC,SIT-DOWN, FRONT WHEEL DRIVE, REAR WHEEL STEER; GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers battery powered, electric, sit-down, industrial warehouse and shipboard type fork lift trucks.

1.2 Classification. Trucks shall be of type E(general), EE(spark enclosed), or EX(explosion proof) and as specified on the applicable military specification sheet.

2. APPLICABLE DOCUMENTS

2.1 Government Documents.

2.1.1 Specifications and standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation, (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to:Commander, Navy Ships Parts Control Center, 5450 Carlisle Pike, P.O. Box 2020, Mechanicsburg, PA 17055, Attn: MHE Management and Engineering Department, Code 036 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N4779 FSC 3930

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SPECIFICATIONS

FEDERAL

W-B-133 - Batteries, Storage(Lead Acid, Industrial for Cycle Service)

QQ-P-416 - Plating, Cadmium (Electrodeposited)

GG-P-455 - Plates and Foils, Photographic (Photosensitive Anodized Aluminum)

TT-E-489 - Enamel, Alkyd, Gloss (for Exterior or Interior Surfaces)

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MIL-V-173 - Varnish, Moisture-and-Fungus-Resistant (for the Treatment of Communication, Electronic, and Associated Electrical Equipment)

MIL-P-514 - Plates, Identification, Instruction, and Marking, Blank

MIL-S-901 - Shock Tests, H.I. (High Impact); Shipboard Machinery, Equipment, and Systems, Requirements for

MIL-L-2104 - Lubricating Oil, Internal Combustion Engine, Tactical Service

MIL-G-3859 - Grease Guns, Hand-Operated, Lever, Push and Screw Type

MIL-M-3971 - Meter, Time Totalizing, Non-Hermetically Sealed, Electric, AC and DC

MIL-W-5044 - Walkway Compound, Nonslip, and Walkway Matting, Nonslip

MIL-H-17672 - Hydraulic Fluid, Petroleum Inhibited

(See supplement 1 for list of associated specifications.)

STANDARDS

FEDERAL

FED-STD-595 - Colors

FED-STD-H28 - Screw Thread Standards for Federal Services

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MIL-STD-129 - Marking for Shipment and Storage

MIL-STD-130 - Identification Marking of U.S. Military Property

MIL-STD-162 - Material Handling Equipment Preparation for Shipment Storage Cyclic Maintenance Routine Testing and Process

MIL-STD-209 - Slinging Eyes and Attachments for Lifting and Tying Down Military Equipment

MIL-STD-461 - Electromagnetic Emission Susceptability Requirements for the Control of Electromagnetic Interference

MIL-STD-810 - Environmental Test Methods

MIL-STD-889 - Dissimilar Metals

MIL-STD-1366 - Material Transportation System Dimensional and Weight Constraints, Definition Of

MIL-STD-1472 - Human Engineering Design Criteria For Military Systems, Equipment and Facilities

MIL-STD-1791 - Designing For Internal Aerial Delivery In Fixed Wing Aircraft (Request For Issue Other Then DOD Must Be Sent Via ASD/ENES, Wright Patterson AFB, OH 45433-6503

MS-15367 - Battery, Storage, for Electric Powered Industrial Trucks and Tractors

MS-16974 - Battery, Storage, Type "EX" for 4,000- and 6,000-Pound Capacity, Fork Lift Trucks

HANDBOOKS

MILITARY

MIL-HDBK-157 - Transportability Criteria

MIL-HDBK-267(SH)- Guide for Selection of Lubricants and Hydraulic Fluids for use in Shipboard Equipment.

AFS DH 1-11 Design Handbook, Air Transportability (Application for copies should be addressed to: ASD/ENESS, Wright Patterson AFB, OH 45433-6503)

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(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (Attn: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DEPARTMENT OF LABOR(DOL)

29 CFR, Part 1910 - Occupational Safety and Health Standards (OSHA)

(Application for copies should be made to a local DOL officer or the Superintendent of Documents , U. S. Government Printing Office, Washington DC 20402)

2.2 Non-Government Publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS are the issue of the documents cited in the solicitation (see 6.2).

ASSOCIATION OF AMERICAN RAILROADS

Outline Diagram for Single Loads, Without End Overhang, On Open-Top Cars (Application for copies should be addressed to the Associaiton of American Railroads, 59 East Van Buren St., Chicago, IL 60605

AMERICAN NATIONAL STANDARDS INSTITUTE(ANSI)

B93.28 - Methods for Calibration of Liquid Particle Counter

(Application for copies should be addressed to: American National Standards Institute, 1430 Broadway, New York, New York 10018).

AMERICAN SOCIETY OF MECHANICAL ENGINEERS(ASME)

B 56.1 - Safety Standard For Low Lift and High Lift Trucks

B 56.11.3 - Load Handling Symbols for Powered Industrial Trucks

B 56.11.4 - Forks and Fork Carriers for Powered Industrial Trucks

(Applications for copies should be addressed to: American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B 633- Electrodeposited Coatings of Zinc on Iron and Steel

(Application for copies should be addressed to: American Society for Testing and Materials, 1961 Race Street, Philadelplhia, PA 19103).

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE J429 - Mechanical and Material Requirements for Externally Threaded Fasteners

SAE J492 - Rivets and Riveting

SAE J514 - Hydraulic Tube Fitting

SAE J516 - Hydraulic Hose Fittings

SAE J517 - Hydraulic Hose

SAE J518 - Hydraulic Flanged Tube, Pipe and Hose Connections, 4-Bolt Split Flange Type

SAE J 524- Seamless Low Carbon Steel Tubing Annealed for Bending and Flaring

SAE J525 - Welded and Cold Drawn Low Carbon Steel Tubing Annealed for Bending and Flaring

SAE J534 - Lubrication Fittings

SAE J536 - Hose Clamps

SAE J551 - Performance Levels and Methods of Measurement of Electromagnetic Radiation from Vehicles and Devices (30-1000 Mhz)

SAE J917 - Marine Push-Pull Control Cables

SAE J925 - Minimum Service Access Dimensions for Off-Road Machines

(Application for copies should be addressed to: Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, Pennsylvania 15096).

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THE TIRE AND RIM ASSOCIATION, INC.(T&RA)

Tire and Rim Association Yearbook

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 W. Market St., Akron, OH 44313)

UNDERWRITERS LABORATORIES, INC.(UL)

UL 583- Standard for Electric-Battery-Powered Industrial Trucks

(Application for copies should be addressed to: Underwriters' Laboratories, Inc. 1285 Walt Whitman Road, Melville, Long Island, New York 11746; 207 East Ohio Street, Chicago. Illinois 60611; or 1655 Scott Boulevard, Santa Clara, California 95050).

(Nongovernment standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of Precedence. In the event of a conflict between the text of this document and the references cited herein,(except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Specification Sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.1.1 First article. When specified (see 6.2), a first produced truck shall be subjected to a first article inspection (see 6.2) in accordance with 4.3.

3.1.2 Material. Material shall be as specified herein. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification. When specified for shipboard use or for resistance to saline atmosphere corrosion (see 6.2), all piston rods, shafts, axles, chains, keys, pins, and other components which must be movable or subject to relative movement during any use of the truck, or which must be removed for service or repair of the truck or its components, shall exhibit no evidence of malfunction or damage to surfaces due to corrosion when tested in accordance with Section 4. Minor corrosion affecting only appearance shall not be sufficient cause for rejection. Asbestos material or radioactive material shall not be used for any end product for this vehicle.

3.1.2.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals and methods of protecting them are defined and described in MIL-STD-889.

3.1.2.2 Plating. Unless otherwise specified (see 6.2), all threaded fasteners, washers and cotter pins required to fabricate the truck, except those in contact with oils in reservoirs and those inside corrosion resist components shall be zinc plated, cadmium plated or made of corrosion resis g material. When specified for shipboard use or for resistance to saline atmosphere (see 6.2) zinc plating shall be in accordance with ASTM B633 type II, SC 3 and cadmium plating shall be in accordance with QQ-P-416, type II, class 2. For all other applications, zinc plating shall be in accordance with ASTM B633, type I, SC 1 and cadmium plating shall be in accordance with QQ-P-416, type I, class 3.

3.1.2.3. Cold Storage Conditioning. When specified (see 6.2) cold storage conditioning is required to compensate for the corrosive effects resultant from condensation of moisture present in cold storage applications. The requirements are for trucks operating in an environment where the temperature fluctuates from normal warehouse ambient temperature to -40°F (-40°C) or for continued operation at temperatures of up to -40°F (-40°C). Sealed bearings shall be used on all pulleys, rollers, and shafts. All lubricants and oils shall be suitable for the intended service as recommended by suppliers. All electrical components subject to water shall be sprayed with an appropriate protective coating. Motors and electrical controls shall be suitable for operation in the specified environment and modified if required to eliminate condensation of moisture and possible corrosion damage.

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All other electrical controls and switches shall be of a design required to provide protection against corrosive damage resulting from the condensation of moisture. All fasteners shall be treated for corrosion resistance. All piston rods, shafts, axles, chains, keys, pins and other components which must be movable or subject to relative movement during any use of the truck, or which must be removed for service or repair of the truck or its components, shall exhibit no evidence of malfunction or damage to surfaces due to corrosion resulting from moisture condensation.

3.1.3 Design. The trucks shall be designed for the specified capacity at the specified load center with the forks at the specified minimum fork height. Design of truck covered by the specification and all components therein shall insure safe operation when operated as specified herein. There shall be no evidence of accelerated wear, failures or permanent deformation when operated as specified herein. There shall be no exposed bolts, clamps, gages, fittings, lifting parts, or appendages that can be caught or hooked while the truck is working in a confined space.

3.1.3.1 Operating temperature. Trucks shall be designed to operate in a nominal ambient temperature of 115°F (46°C) (see 6.2).

3.1.3.2 Safety. The truck shall conform to the applicable requirements of UL 583 for type E trucks, type EE trucks, and type EX trucks, such additional requirements as specified herein, and with the Occupational Safety and Health Standards requirements of 29 CFR, Chapter XVII, Part 1910. All rotating and reciprocating parts of the forklift, excluding the mast, shall be enclosed or guarded.

3.1.3.2.1 Certification. Trucks shall bear U. L. stamp of approval for type E, type EE, or type EX as specified. In addition, the contractor shall submit to the contracting officer or his authorized representative satisfatory evidence that the trucks furnished under this specification meet other requirements of 3.1.3.2. Acceptable evidence of meeting these other requirements shall be certified test reports from recognized independent testing laboratories acceptable to the Government, indicating that the trucks conform to the requirements of 3.1.3.2. Also, the truck shall be furnished with a safety nameplate specified herein (see 3.10.4).

3.1.3.3 Bearings. Unless otherwise specified herein, rotating parts shall be mounted on sealed ball, roller or tapered bearings.

3.1.3.4 Lubrication. All surfaces requiring lubrication shall be provided with a suitable means for lubricating except where sealed bearings are used.

3.1.3.4.1 Lubricants. The truck shall operate as specified herein when lubricated with standard lubricants except that shipboard trucks shall only use lubricants listed in MIL-HDBK-267(SH).

3.1.3.4.2 Lubrication fittings. Lubrication fittings shall conform to SAE J534. Fittings shall be located in a protected position and shall be accessible to a hand-operated grease gun conforming to MIL-G-3859 with a 10-inch (250mm) flexible extension.

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Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Panel and plates equipped with hand-operable, quick-disconnect fasteners or no more than eight common fasteners may be removed to provide accessibility.

3.1.3.4.3 Lubrication seals. All lubrication seals shall be properly sized and seated to prevent leakage of lubricant through or around positioned seals.

3.1.3.5 Screw threads. Screw threads shall be in accordance with FED-STD-H28.

3.1.3.6 Bumper strip. All EX trucks shall be provided with bumper strip in accordance with UL 583. The bumper strip shall provide complete bumper protection on both sides and the rear of the vehicle. It shall be so located on the vehicle that it shall prevent any portion of the battery or truck, including towing eye, from making contact with any other surface except through the bumper strip.

3.1.3.7 Maintainability. Provision shall be made for adjustment, servicing, and replacement of all electrical assemblies and components, battery, wearing parts of lift and tilt mechanism, brakes and components, wearing parts of steering assembly, tires, wheels, lights, and horn, without the removal of any major parts other than covers. If hand access openings are used on the exterior of the truck, the edges of each opening shall be smooth and shall be provided with a removable or hinged cover. Dimensions of access openings shall be in accordance with SAE J925 for bare hand or normally clothed man. There shall be no interference with the servicing or draining of oils and lubricants to or from any assembly or components by frame members or other obstructions, and without requiring the use of flexible connections, except as specified in 3.1.3.4.2, or other special devices. Drain holes with removable drain plugs shall provide for complete drainage without disconnection of hoses. Compliance with specific maintenance requirement for the equipment shall be t ed in accordance with Section 4. Unless otherwise specified (see 6.2), all special hand tools required for maintenance of the truck shall be provided and shipped with each truck procured. The truck shall be designed so as not to require preventive maintenance or repair services at intervals shorter than 2000 laps when tested as specified herein except for servicing truck batteries and removing foreigh matter from tire threads. Reservoir level checks are not considered scheduled maintenance. The ratio of man hours of scheduled and unscheduled maintenance required to the hours of operation performed shall not exceed .08.

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3.1.3.7.1 Maintenance operations. Unless otherwise specified herein, each operation in the following maintenance operations list shall be accomplished by one man in not more than the time specified, using common tools and special tools, designed for the specific purpose of truck maintenance ,if any,furnished with the truck.

Maintenance Operation (Time in Minutes)	E	EE	EX
a. Remove and Replace Drive Motor Brushes	60	90	150
b. Remove and Replace Hydraulic Pump Motor Brushes	60	90	150
c. Remove and Replace Steer Motor Brushes	60	90	150
d. Remove and Replace all Contactor Tips	60	90	150
e. Remove and Replace all Filters, Screens and Strainers in Hydraulic System	105	105	105
f. Bleed and Adjust Brakes and Refill Master Cylinder (2 men) (4-wheel brakes excepted)	45	45	45

3.1.3.8 Lubrication system. When specified (see 6.2), a lubrication system described as follows shall be provided. It shall include:

(a) A system for lubrication of all points that require service at intervals of not less than 2000 laps when tested as specified herein.

(b) A system not requiring pumping, metering device, or electrical controls. The system shall be limited to the extension of individual lubrication points; the design of joints, bearings and seals to withstand the prolonged lubrication cycle; and accessibility to lubrication points. Neither shall it be necessary to remove the battery for access to lubrication points.

(c) All necessary lubrication points capable of being serviced with the machine at rest in normal mode on a level warehouse floor. Exceptions: procedure may incorporate operation of the mast or carriage assembly and movement of steering mechanism.

(d) It shall not be necessary to remove components or assemblies to facilitate lubrication. However, this does not preclude the removal of nonstructural covers, provided that such covers are accessible within parameter (a) above and are removable by use of hand tools and manual labor only. The time required for removal and replacement of all covers shall not exceed one-tenth manhour total per complete lubrication service.

(e) Each point requiring lubrication shall be lubricated separately. Non-accessible lubrication points may be extended by flexible or rigid grease lines or hoses as necessary to points of access. However, all lubrication system parts, separately and in aggregate, shall be constructed and located so that operational capability of the end item will not be degraded.

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(f) The location of hoses, lines and lubrication points to the design outline of the truck shall be so designed to provide for protection of hoses and lines from sagging, dragging, or entanglement with foreign objects during operation.

3.1.4 Electrical equipment.

3.1.4.1 Resistors. Resistors shall not be used in series with the motor armature for speed control. Speed control shall be effected by the application of a system as specified in 3.2.1.4. Low resistance shunts across field coils may be used to increase top speed of the motor. Such resistors as are necessary in the control circuit network shall be continuous-duty types of a power rating value sufficient to withstand all tests required by this specification.

3.1.4.2 Enclosure and shields. Trucks shall comply with the requirements of UL 583 except motor enclosures of shipboard trucks in minimum 115°F (46°C) ambient shall not exceed 250°F (121°C) when subjected to any test specified herein. All electrical components including wiring shall be so located or protected by adequate means to prevent contact by excess moisture or battery electrolyte in the event of spillage.

3.1.4.3 Overcurrent devices. In addition to the requirements of 3.1.3.2, overcurrent devices shall be provided on each power and each control circuit.

3.1.4.4 Heat-producing devices - Type EE and Type EX trucks. The outside surface temperature of heat-producing devices shall stabilize at temperatures allowed by UL 583 when truck is operating with rated capacity continuously in the speed which creates the most heat, except that heat producing devices of EE trucks for shipboard use shall not exceed 250°F (121°C) and heat producing devices on EX shipboard trucks shall not exceed 225°F (107°C).

3.1.4.5 Static discharge device. Type EE and EX trucks shall be equipped with static discharge device in accordance with UL 583.

3.1.4.6 Control panel. The control panel shall be designed for removal as a unit or front removal of electrical assemblies mounted thereon. Mounting studs, bolts, or nuts, shall be so secured that they will not turn or become loose when an assembly is removed from the panel. The control panel cover shall permit removal of the cover for inspection and servicing without removal of any major part or interference from truck structure.

3.1.4.7 Conductors. Conductors and enclosures shall comply with all requirements of UL 583. All wiring shall be color or number coded.

3.2 Propulsion and hydraulic systems.

3.2.1 Motor and motor accessories.

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3.2.1.1 Motors - Type E trucks. Traction motors shall be industrial type, with a high starting torque, designed for operation on 36 or 48(see 6.2) direct current voltage. Motors shall be capable of withstanding the stresses and current loads induced by test as required by this specification. The resultant strains and temperature rise shall not cause failure of any component part or excessive pitting other than normal discoloration of the commutators. The hydraulic pump and steering pump motor, if used, shall successfully withstand the stresses and current loads induced by tests required by this specification. All electric motors shall have bolt-on conductor connections.

3.2.1.2 Motors - Type EE and EX trucks. In addition to the requirements of 3.2.1.1, motor(s) for type EE and EX trucks shall be provided with one or more thermal relay switches which shall cause disconnection of battery power to the affected circuit before the outside surface temperature of the motor(s) exceed limits as specified in 3.1.4.4. Thermal relays shall be located in critical heat areas and shall interrupt the current only to the over-heated circuit.

3.2.1.3 Insulation. Insulation for motors shall not contain silicone materials. Silicone material in any form shall not be used in the motor. The electrical and mechanical properties of the insulated windings shall not be impaired by the application of the temperature permitted for the class insulation used (the word "impaired" is here used in the same sense of causing any change which could disqualify the insulating material for continuous service).

3.2.1.4 Speed controller. The speed control for type E and EE shall be solid state type. The controls shall be designed for continuous use and shall not include failure or damage when the truck is tested in accordance with Section 4. The speed controller for type EX trucks shall provide stepless, 4-step, or combination step and stepless control in forward and reverse directions. If either 4-step or combination step and stepless control is used, a time delay stepping mechanism shall be used. Resistors shall not be used for speed control of the truck. Low resistance shunt field controls may be used to increase the speed of the truck. The controller shall provide means to permit precision maneuvering in restricted areas. It shall provide stepless or combination step and stepless acceleration through the complete speed range from zero to full speed in both the forward and reverse directions and shall have automatic electrical or mechanical interlocks to prevent starting except in low speed. It shall provide automatic means of dynamic braking to limit the plugging when the controller is reversed with the truck in motion. All contactors on a given truck shall be designed to provide maximum interchangeability of parts. Contactors shall be designed for operation on 36 volts, or when specified (see 6.2) 48 volts, direct current and shall have easily replaceable tips of copper or silver alloy composition. Replacement of contactor tips shall be accomplished by use of hand tools only. There shall be no evidence of failure in any part of the controller when the truck undergoes tests specified herein.

3.2.1.5 Safety cutout system.

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3.2.1.5.1 Type E truck. Provisions shall be made for type E trucks that all power to the traction motor is interrupted when the operator leaves his seat.

3.2.1.5.2 Type EE and EX trucks. In order to insure that no current is flowing when the connector is being disconnected on type EE and EX trucks, a magnetic contactor shall be installed in one of the two main power lines from the battery connector. The control circuit for the magnetic contactor shall be wired with two switches in series, either of which shall cause the contactor to open and disconnect all power. The first switch shall be a seat switch. The second switch shall be an emergency switch with a positively attached mushroom type button located within easy reach of the operator. For type EX trucks the emergency switch shall be an enclosure to comply with class I group D and class II group G of UL 583.

3.2.2. Battery and battery accessories. Unless otherwise specified (see 6.2), battery size and capacity for type E and EE trucks shall be governed by MS-15367. Unless otherwise specified (see 6.2), battery leads for type E and EE trucks shall be equipped with a type EC locking half connector, that is used to connect with electric motor through mounting half or to connect to battery charger with charging half. When battery well is provided, the length of leads shall be 18 (460 mm) inches in lieu of 30 inches (760 mm) specified in MS-15367. Batteries for type EX trucks shall be as specified by MS-16974. When furnished by truck supplier (see 6.2), the battery, battery leads, and connectors on type EE and type EX trucks shall be shielded or modified as necessary to assure that the battery as a component of the truck when drawing maximum current (lifting and traveling simultaneously) shall comply with 3.9 as applicable. When first produced truck is specified (see 6.2), the battery for the first produced truck used for testing by the contractor, if supplied, shall be shipped in a wet and charged condition after the exterior of the battery has been rehabilitated to a "like new" appearance.

3.2.3 Battery connector mounting. Unless otherwise specified (see 6.2), type E and type EE trucks shall be equipped with Standard EC mounting half connectors, suitably located to fit the locking half as specified in 3.2.2. Type EX trucks shall have Type N three contact plug on leads to mate with the charging receptacle specified in MS-16974.

3.2.3.1 Battery well. Battery well dimensions shall be determined from batteries specified in 3.2.2. The battery well side plates shall be approximately 5.5 inches (140 mm) high. The well construction shall be in accordance with UL 583. The inside of the well shall be finished with an acid resistant paint to resist deterioration through normal contact with electrolyte.

3.2.3.2 Drive unit. Each gear, with the exception of worm gears, shaft and axle shall be of heat-treated alloy steel. The drive shall be a gear reduction unit(s) connected to the driving motor(s), totally enclosed and operating in a constant oil bath. The final gear reduction may be in a totally enclosed casing but be arranged for positive periodic lubrication of gear teeth. Drive gear housing(s) shall be equipped with filler and magnetic drain plug.

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Provision shall be made for adjustment where necessary to compensate for wear in the differential and pinion. All required drain plugs, check plugs, or oil level checking and filling devices shall be accessible without removing major components.

3.2.4 Hydraulic system. The hydraulic system shall consist of all hydraulic components necessary for operation of the forks, mast, and steering and does not include brake components. The system shall have a maximum working pressure (relief valve setting) not in excess of 2900 psi (20 MPa). Oil for the system shall conform to MIL-L-2104, Grade 10, MIL-H-17672, or commercial hydraulic fluid with rust and oxidization inhibitor, anti-foaming, anti-wear, turbine grade base. Hydraulic fluid initially shall not be higher in contamination level than will pass through the filter required herein. All hydraulic component materials shall be compatible with these oils. Unless otherwise specified herein, all system components which are subject to full system pressure shall withstand a proof pressure that is equal to or greater than 1.5 times the maximum working pressure (1.5 times relief valve setting) without external leakage, damage, or permanent deformation. Hydraulic fittings on pumps, valves, and cylinders shall be the O-ring straight thread type as specified in SAE J514, 4-bolt flanged type as specified in SAE J518, or pipe thread type. The hydraulic system shall be cleaned free of foreign matter including weld slag and spatter. Hydraulic connection ports of the hydraulic system components shall be sealed by the component manufacturer. The system shall provide for lowering of rated load at the rate specified in 3.6.2 in the event of failure of or damage to hydraulic hose supplying the lift cylinder. A pipe tap or straight thread O-ring boss test point shall be located at the outlet of each steering pump or pressure inlet of each control valve for measuring pressure.

3.2.4.1 Hydraulic lines. Hydraulic lines shall consist of tubing, hose and fittings. All lines shall be routed to provide the shortest line, minimum number of bends, ease of maintenance and maximum protection. Provision shall be made to prevent damage of lines due to chafing. The hose installation, such as angle of fittings and location, shall cause no stress concentration on the hose at the fitting, and the hose shall be supported, when necessary, to eliminate sagging at the fitting. All lines other than suction lines, return lines, and cylinder bleeder lines shall be considered pressure lines. Copper tubing shall not be used in the brake system.

3.2.4.1.1 Pressure hose assemblies. Pressure hose shall be as specified in SAE J517, except 100R7 type hose is not acceptable for shipboard trucks. Permanently attached type pressure hose couplings are acceptable as original equipment hose assemblies. For shipboard trucks field attachable reusable fittings shall be compatible with the truck design such that they will replace each permanently attached coupling, and the replacement hose assembly shall fit in place of the original hose. All fittings shall be either 37-degree flare, female swivel or 4-bolt split flange as specified in SAE J516. The hose installation shall not cause the hose to be bent beyond the minimum bend radius specified in SAE J517.

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3.2.4.1.2 Pressure tubing. Pressure tubing shall be in accordance with SAE J524 or SAE J525 and shall meet the requirements specified herein. Tubing shall have a minimum burst pressure that is equal to or greater than 3.5 times the system maximum working pressure. Tubing shall be clamped by cushioned, threaded fasteners spaced at intervals of not more than 24 inches (610 mm). All bends shall be smooth without flattening, kinking, or wrinkling of the tube. The term "without flattening" is defined as follows:

Throughout the bend, the tube maximum diameter shall be equal to or greater than the tube nominal diameter, and the tube maximum diameter minus the tube minimum diameter shall be not greater than 15 percent of the tube nominal diameter.

3.2.4.1.3 Pressure tube fittings. Unless otherwise specified herein, pressure tube fittings shall be 37-degree flare or "O" ring boss conforming to SAE J514 or 4-bolt split flange conforming to SAE J518. Jump size tees may be used. Four-bolt split flange connections may terminate in either a flange head that incorporates an SAE J518 mating face or a flange head that integrally incorporates 4-bolt holes and an "O" ring groove for mating to a SAE J518 port face. Pressure tube fittings that incorporate the SAE J518 connection shall contain a boss into which the tube shall be brazed or welded. Pressure tube fittings shall have a minimum burst pressure that is equal to or greater than 3.5 times the maximum system working pressure.

3.2.4.1.4 Suction line. The pump suction line (pump inlet line) shall include provision for flexibility when there is vibration or movement between the reservoir and the pump. The suction line shall be sized so that the minimum pump inlet pressure does not fall below atmosphere by more than 5 inches (130 mm) of mercury at 150°F (66°C), plus or minus 10°F (5°C), oil temperature. The suction line installation shall include no stress concentration at the end fittings and no kinks. Suction hose shall conform to SAE J517, 100R4 or shall be oil resistant hose. The suction line assembly shall be capable of withstanding a minimum of 25 inches (84 KPa) of mercury vacuum without evidence of leakage or collapse. Suction hose clamps shall be of the double wrap stainless steel type in accordance with SAE J536, Type F, Style 1, 2, or 4.

3.2.4.2 Hydraulic pump(s). The hydraulic pump(s) shall withstand a minimum proof pressure of 1.5 times the maximum system working pressure without permanent deformation, damage, or external leakage.

3.2.4.3 System filtration. A system filter shall be furnished in either the suction or return line as specified in 3.2.4.3.1 or 3.2.4.3.2.

3.2.4.3.1 Return line filter. The filter shall maintain a full flow condition for the duration of reliability testing as specified in 4.3.2.31. The filter bypass may open for system oil at a temperature less than 70°F (21°C).

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All return oil shall pass through the filter except relief oil which may return directly to the reservoir. The filter shall be located so that the element may be removed for examination or replacement without spilling or draining oil from the reservoir. At the maximum system oil flow through the filter, including lowering rated load on the forks and an oil temperature of 150°F (66°C), plus or minus 5°F (3°C), the pressure drop across a new filter assembly shall not exceed 10 psi (69 KPa). An automatic bypass shall be provided in the filter and shall open at not less than 10 psi (69 KPa) greater than pressure drop measured across new filter assembly and shall permit full flow at a pressure not greater than 35 psi (241 KPa). The filter housing shall have a minimum proof pressure of 100 psi (690 KPa) without permanent deformation, damage, or leakage except when the hydraulic reservoir serves as filter housing. The filter shall be equipped with a contamination indicator which may be remotely mounted. The indicator shall be clearly readable without the removal of any components. As a minimum, the indicator shall show when the element is clean and when the filter is beginning to bypass, requiring a change of the element.

3.2.4.3.2 Suction filter. The suction filter shall conform to 3.2.4.3.1 with the following exceptions:

(a) Pressure drop across a new filter assembly shall not exceed 0.75 psi (5.2 KPa).

(b) The automatic bypass shall open at not less than 2.0 psi (14 KPa) and shall permit full flow at not more than 3.5 psi (24.1 KPa).

(c) The filter housing shall have a proof pressure of not less than 30 psi (207 KPa).

3.2.4.4 Reservoir. A reservoir shall be furnished to contain the hydraulic system oil. The reservoir shall include a filler opening, screen, cap, and other items required to function as specified herein. The reservoir shall be rigidly mounted to prevent vibrations that could result in leaks, loosening of fasteners, brinelling of material, or cracks. The location shall not impair operational visibility or access to either side of the operator's compartment. The reservoir shall have sufficient capacity to prevent air entering the system with all hydraulic pistons fully extended and sufficient free air capacity to prevent oil being discharged through the reservoir air vent when maximum return flow of oil is surged into the reservoir from the system. The "Add" and "Full" marks on the dipstick shall indicate the correct oil level when the truck is on level ground, the mast is vertical, and the forks are at ground level. The filler cap and tube shall be located to provide for filling from a standard 5-gallon (19 L) container conforming to MIL-C-1283 and to prevent entrance of contaminants throughout filling and checking operations. The filler tube shall be not less than 1 inch (25 mm) inside diameter.

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The reservoir pump suction openings shall be located a minimum of 0.375 inch (10 mm) above the bottom of the reservoir to prevent entrance of settled contaminants. Except for sealed reservoirs, a breather with a maximum rating of 15 micron nominal shall be provided and shall filter incoming air. The breather shall preclude the entrance of water induced by rain. All return line flow except vent line shall discharge into the reservoir at a level which is below the oil level in the reservoir when the lift cylinder is at maximum extension. The reservoir filler neck shall contain a removable 30 mesh strainer retained in place unless all oil added to the reservoir is filtered by an equal or finer filter prior to passing through the pump. The reservoir shall have one or more accesses for manual cleaning of the entire inside or be removable for cleaning. A removable strainer of 50 or less mesh openings or a removable filter shall be installed in either the hydraulic reservoir or the suction line to protect the pump. When a strainer is provided for pump protection, a removable filter must be installed in the hydraulic system. For type EE and EX trucks, two or more thermal switches shall be installed in different critical heat areas on the same hydraulic reservoir and each one, separately, shall cause disconnection of the pump motor(s) before the temperature rises above that specified in 3.1.4.4.

3.2.4.5 Hydraulic control valves. Mounting of the valves shall be rigid and shall permit no vibration that may cause instability of the controls. All control-valve spools shall be spring centered with sufficient force to return the levers to the center position and retain them there throughout all operations. Valve spools exposed to dirt and moisture shall be provided with wipers and seals. Spools shall not bind under any operating conditions specified herein.

3.2.4.6 Cylinders. All hydraulic system and steering cylinders shall be fabricated from either seamless steel tubing or welded and redrawn tubing. All welding and brazing to the cylinder within the zone of piston operation shall be done prior to final machine honing or rolling of the cylinder. Cylinders shall be so located as to provide for ease of maintenance and replacement. All internal cylinder ports shall be located beyond the area of piston travel. Each cylinder shall have an exclusion device to prevent foreign material and fluids from entering around the piston rod and damaging the seal and other cylinder components. The rod shall be hard chrome plated to prevent corrosion. After plating, maximum rod surface roughness shall not exceed 32 micro-inches (0.8 micrometer). All hydraulic system and steering cylinders shall withstand a pressure that is equal to 1.5 times the system relief valve setting without evidence of rupture, permanent deformation, damage, or external leakage. The bearing surface shall be of a material that will not scar, score, or gall the inside surface of the cylinder or the cylinder rod.

3.2.4.7 Hose reels. When hydraulic hose reels are used on the mast to maintain hose tension during mast extension, sheave wheels, guide rollers, guide blocks, or other means shall be provided to prevent the hose from becoming entangled around and riding over the reels or reel flanges. Hose reels shall be mounted such that they are within the plan outline of the truck for all positions of the forks.

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3.2.4.8 Contamination levels for hydraulic system. When the hydraulic system is tested as specified in 4.3.2.33, particle counts shall not exceed 2000 particles per milliliter greater than 10 micrometers, and particle counts shall not exceed 20 particles per milliliter greater than 20 micrometers.

3.3 Structure.

3.3.1 Chassis and frame. The frame and related structure shall be fabricated from steel and steel castings capable of withstanding 300 percent of rated load without permanent deformation. The frame and related structure shall have sufficient depth to afford protection for working parts and shall be capable of withstanding,without permanent deformation, stresses induced by capacity loads and tests in Section 4. When specified (see 6.2), the complete frame structure shall be tested as outlined in 4.3.2.27. The width of the truck shall not exceed the dimension specified on the applicable specification sheet.

3.3.1.1 Uprights and carriage assembly. Uprights shall be of the telescopic roller type. Rollers shall be of the permanently lubricated for life type. Uprights shall be positioned in such a manner that the back surface of the lower fork mounting plate or cross bars in the vicinity of the tires are located forward of the vertical plane tangent with the front face of the drive tires. The upright shall have two lift cylinders ,each located near or within the upright chenels, instead of one lift cylinder located in the middle of the mast except one free lift cylinder which is no higher than tht top of the carriage is permitted. The lift cylinders shall be mounted and positioned so that they do not protrude forward of the plan outline of the uprights. Chains shall be installed in such a manner that they can not be displaced from the roller until they have been raised a minimum of 0.375 inch (10 mm) or a guard that prevents the chain from coming off the sheave is removed. Cross members of the mast structure, when in a lowered position, shall not interfere with the operator's horizontal line of sight plus or minus 10 degrees to the front of the truck. The operator shall be able to see the tip of at least one fork at any lift height without leaving his seat when there is no load on the forks. The lift shall be hydraulically actuated, with the upper portion automatically lifting as the load is raised above the "free lift" position. Means shall be provided in the uprights to compensate by replacement or adjustment for wear which may develop between lateral rolling contacts. Means shall be provided to prevent overtravel of the carriage or channel in both upper and lower positions. Chains used in the lifting mechanism shall have a factor of safety of not less than 4.6 to 1 based on average ultimate strength. When specified (see 6.2), load securing eyes or rings shall be provided on both the left and right sides of the carriage assembly in accordance with MIL-STD-209. The load securing devices, when not in use shall not extend forward of the surface on the front face of the carriage. The load securing devices shall be capable of withstanding a horizontal force equal to the truck's capacity without failure or deformation. Structural safety factor for the load securing devices shall not be less than 3 to 1 based on yield strength, assuming each devie is loaded equally.

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3.3.1.2 Forks. The forks and fork carrier shall conform to ASME B56.1, ASME B56.11.4 and the applicable specification sheet. A cutout shall be provided in the fork carrier so that the lower hook of the fork will fit through the slot and each fork shall be removable without removing the load backrest. Fork thickness shall not exceed 0.5 inch (13 mm) at the tips and the end taper shall be not less than 14 inches (355 mm) long. Taper shall be on the underside of the forks. The dimensions and spacing adjustment shall be measured between outer edges of the forks. The top of the fork tips shall be chamfered or rounded to break corners and provide smooth edges. The fork carrier shall be free of obstruction to allow the forks to slide over the entire width of the carrier. Unless otherwise specified, (see 6.2), forks on trucks up to but not including 10,000-pound (4540 kg) capacity shall be adjustable laterally without the use of hand tools. Unless otherwise specified, (see 6.2), trucks of 10,000 pounds (4540 kg) capacity shall provide for adjustment of the unloaded forks simultaneously by auxiliary means, i.e., crank and screw or similar device, from one side of the fork carriage by one man without undue effort, and the adjusting mechanism shall not bind or break under applicable test of Section 4. Undue effort shall be defined as more than 20 pounds (89 N) of force at the handle. Protruding handles and levers shall fold downward or inward in order to eliminate interference with the truck operation. The adjusting mechanism shall be protected to prevent foreign matter such as dirt and mud from collecting on its surfaces. When specified (see 6.2), forks shall be hydraulically adjustable (unloaded) from the operator's position. Each fork arm shall be stamped with its rated capacity in a location clearly discernible and not subject to wear. When shipboard trucks are specified (see 6.2), folding forks shall be furnished. Folding forks shall meet all the requirements herein, including the overload test in Section 4. Each folding fork shall be hinged or pivoted near the heel and shall be capable of being folded back manually against the fork upright and the backrest if applicable. A manually operated locking device shall hold the fork in the upright position. The locking device shall not interfere with the operation of the truck when the forks are down. The forks on EX trucks shall be safeguarded against mechanical sparking in accordance with ANSI/UL 583.

3.3.1.3 Load backrest extension. A metal load backrest extension, removable without disturbing any other component, shall be provided. There shall be no protruding bolts or appendages beyond the side plane of the load backrest. The load backrest extension in conjunction with fork and hanger design shall provide a vertical rear guard at least 48 inches (1220 mm) high measured from load carrying surfaces of the forks. It shall have width and capacity and strength sufficient to prevent a rated load 48 inches (1220 mm) high or less from forcing backrest against the uprights when the mast is at maximum rearward tilt. The rack shall be firmly attached to the lift carriage. Rack design shall not obstruct operator's visibility. The spacing of vertical members shall be no more than 6 inches (150 mm) between members. The backrest shall be capable of withstanding without permanent deformation a horizontal static test load equal to the rated load uniformly distributed over the area, from the top carrying surface of the forks to the top of the backrest for not less than 3 minutes. Load backrest extension for type EX truck shall comply with requirements of ANSI/UL 583. A positive means or stops shall be provided to prevent the forks from sliding off the fork carriage whenever the load backrest is removed.

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3.3.2. Wheels and Tires

3.3.2.1 Wheels for solid tires. The truck's solid tires shall be mounted on flat-faced steel, malleable iron, or cast iron wheels. There shall be two single front drive wheels and at least two single rear steerable wheels. Wheels, when equipped with new tires, shall clear any part of the truck structure and mast structure by at least 0.25 inch (6 mm) under all conditions of operation.

3.3.2.2 Solid Tires. The tires shall be new and unused and shall be selected from sizes listed in the Tire and Rim Association (T&RA) Yearbook. The tires shall be of the industrial, solid rubber or cushion type, molded onto a new press-on type steel band. Tire loadings shall not exceed the values listed in the T&RA Yearbook. Front tires shall have commercial, non-directional treads, and smooth tires are unacceptable. Rear tires may have either smooth or non-directional type treads. When 4-wheel brakes are required, both front and rear tires shall be lug type, non-directional treads and smooth tires are unacceptable. Tire loadings shall not exceed the values listed in the T&RA Yearbook when tested in accordance with 4.3.2.18. When siped tires are specified on the applicable specification sheet, there shall be two sipes on each lug of the steer tires. When truck is specified for shipboard use (see 6.2), the tires for the drive wheels shall be 18 x 8 x 12.125 inches (460 x 200 x 300 mm) minimum; the tires for the trail wheels shall be 16 x 6 x 10.5 inches (405 x 150 x 265 mm) minimum.

3.3.2.3 Wheels for pneumatic tires. The truck's pneumatic tires shall be mounted on steel, malleable iron, or cast iron wheels. Each wheel shall be mounted on wheel studs or cap screws with self-centering tapered ends or on tapered bolt holes which shall provide a positive method of locating the wheel on the hub. Wheels shall have demountable type rims which can be removed from hubs without deflating tires. Steer wheels shall clear any part of the truck structure by at least 0.375 inch (10 mm). Drive wheel clearance to any part of the truck shall be sufficient to permit the use of standard automotive tire chains; dual drive wheels are required to have clearance for chains on outside wheels and tires only. Spacings between rims of dual tired trucks shall prevent contact between tires when truck is on level surface with rated load.

3.3.2.4 Pneumatic tires. The tires shall be new and unused and shall be selected from the sizes listed in the T&RA Yearbook. The tires shall be pneumatic, high pressure,tube type or tubeless, furnished with non-directional tread and smooth tires are unacceptable. Tire loadings shall not exceed the values listed in the T&RA Yearbook when tested in accordance with 4.3.2.18.

3.3.2.5 Tubes. When tube type tires are furnished, tubes shall be heavy duty type and the size required to fit the tires furnished. Valve stems shall be arranged to permit inflation of mounted tires from the side of the truck.

3.3.3 Steering. An automotive type steering wheel not to exceed 20 inches (510 mm) in diameter, mounted on the steering column, shall turn the steer wheels. Wheels shall steer true in accordance with the Ackermann steering principle.

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Clockwise rotation of the steering wheel shall provide for right-hand turning of the truck while in forward motion. Steering force shall be in accordance with 3.6.4. When power steering is specified (see 6.2), a hydraulic steering booster or full power steer shall be supplied. Provision shall be made for precision adjustment for wear and alignment of all major steering components and for lubrication of all friction points by accessible grease fittings or a lubricant reservoir or shall have permanently lubricated connections. When provided, the reservoir shall be accessible for filling and checking of lubricant level. Means shall be provided to prevent wheels or tires or steering linkage from contacting any part of truck structure, other than stops, irrespective of obstacle magnitude or cramped steer angle. Spindles and kingpins, if used, shall be of heat treated alloy steel.

3.3.4 Brakes.

3.3.4.1 Service brakes. The service brakes shall retard, smoothly control, and stop the truck, both with and without load, under all operating conditions specified herein without fade, noise, uneven braking, or malfunction. All brake system components shall be protected from damage or malfunction from road hazards and shall provide maximum safety, efficiency, and ease of maintenance. The service brake system shall include driver's pedal or treadle, actuator mechanisms, two front-wheel brakes, and all other components required for a complete, matched-brake system based on the gross weight of the truck. The brake foot control shall be covered with a replaceable rubber cover. The brake system shall withstand an operator's brake application force of at least 250 pounds (1110 Newtons) at the pedal or treadle without yielding. The wheel brake assembly rated torque capacity heat rejection characteristics, and installation shall be in accordance with the recommendations of the wheel brake assembly manufacturer. The fixed assemblies shall be positively indexed to the axle assembly. The orientation of hydraulic wheel cylinders shall permit bleeding of all air in the system through the bleeders. Brake lining for shoes or pads shall be of the molded non-asbestos material type intended for heavy duty service and shall be securely bonded to the brake components. Brake lining shall have at least 80 percent of the total lining rubbing surface in contact with the wearing surface during the service life of the lining. Automatic or manual brake adjustment shall be provided. Automatic adjustment shall maintain correct lining to drum or disc adjustment without looseness or lockup. The automatic adjusters shall be of the guarded or enclosed type, protected from environmental damage or malfunction. Manual brake adjustment mechanisms shall be accessible and shall be adjustable using common handtools and a star wheel brake tool. The brake system shall conform to the requirements of ANSI B56.1. When specified on the applicable specification sheet, trucks shall be equipped with brakes mounted on both the drive and steer wheels. Trucks having four wheel brakes shall be equipped with a dual output master cylinder providing a dual hydraulic circuit (split hydraulic) brake system.

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3.3.4.2 Parking brake. A seat-operated parking brake shall be provided. The parking brake shall be an independent mechanical friction brake mounted on the drive motor or drive or propeller shaft rotating with the wheels or a separate mechanical linkage actuating the service brake, capable of holding the truck with rated load on the slope specified for ascension on the applicable specification sheet. When specified (see 6.2), only the parking brake with linkage actuating the service brake shall be provided. The brake shall be automatically activated when the operator leaves the operator's seat and hold the loaded vehicle on the specified slope. When specified (see 6.2), a hand operated parking brake lever with a locking device shall be provided in addition to the seat brake. This manual control shall be located within easy reach of the operator and in a position permitting easy and safe movement on and off the vehicle from either side. If hand lever is of the ratchet type, the ratchet and pawl shall be of heat treated steel. Adjustment of parking brake shoes or band shall be accomplished without the removal of any major assembly other than tires and wheels. This adjustment shall provide positive means to prevent loss of adjustment through inadvertent disengagement of adjusting device.

3.3.5 Body. The truck body (other than counterweight) shall be constructed of steel sheet or plate. If the horizontal floor plate is more than 26 inches (660 mm) above ground level, an intermediate step shall be provided approximately midway and shall not extend beyond the truck profile. The step shall be large enough to accommodate the foot of the 95 percentile man with an overshoe.

3.3.5.1 Seat. A cushion seat and a cushion backrest, adjustable forward and reverse, shall provide comfortable seating for the operator when operating the truck. The seat shall have a minimum adjustment of 4 inches (100 mm) which shall be adjustable by the operator from the operating position without the use of tools. Effective dimension measurement point on seat shall be the lowest point on the seat cushion in the depressed condition with a 150-pound (68 kg) operator seated in the driving position. The seat cushion and backrest shall be of foam rubber or polyurethane covered with slip coated vinyl upholstery having a minimum weight of 40 ounces per yard (1.3 kg per meter) for 54-inch (1370 mm) width.

3.3.5.2 Wheel guards. If drive or steer wheels, when in a straight-ahead position protrude beyond the body of the truck in excess of 1.5 inches (38 mm), suitable wheel guards shall be provided to prevent foreign objects being discharged in the direction of the operator, endangering safety of personnel, or otherwise creating a hazard.

3.3.5.3 Towing hook. A pintle or pin type towing hook shall be provided. It shall be recessed in the center rear of the truck, or shall not protrude beyond the path followed by the rear end of the truck when making a turn of minimum turning radius. Towing hook on EX trucks, when not in use, shall not protrude beyond the bumper strip. The towing hook shall be capable of withstanding without permanent deformation a towing force induced in accordance with 4.3.2.14.

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3.3.5.4 Slinging and tiedown provisions. When specified (see 6.2), permanently affixed slinging and tiedown provisions that enable the truck to be lifted in its normal travel position and to be fastened to the floor or deck of a transportation medium shall be provided. Provisions shall conform to MIL-STD-209, Classes 1 and 2 or Class 3 for type II or III equipment, excluding air transport requirements. Attachments shall be indicated on a transportation plate conforming to MIL-P-514, and slinging and tiedown attachments shall be identified by stenciling or other suitable markings in appropriate locations on the exterior of the equipment. Slinging and tiedown markings shall clearly indicate the purpose of each attachment.

3.3.5.5 Tiedown attachments for HI Shock. When specified to meet the requirements of Grade A item in Type A HI (High-Impact) Shock as specified in MIL-S-901 (see 6.2), the truck shall be furnished with tiedown attachments designed in accordance with the instructions of the truck supplier for securing the unloaded truck in stowed position.

3.3.5.6 Sideshift attachment. When specified (see 6.2), truck shall be equipped with a sideshift attachment. Sideshifter shall be hydraulically controlled and shall provide for a minimum movement of 4 inches (100 mm) +/- 0.25 inches (6mm) on each side of center with rated load. Fork hanger shall permit the side adjustment of forks and shall permit the addition of one extra fork to provide 3 forks. Controls for the sideshift mechanism shall be in accordance with 3.4.6.

3.3.5.7 Weighing device. When specifiec (see 6.2) the truck shall be equipped with a weighing device to measure the amount of weight on the forks. The weighing device shall be connected to and operate from the system for raising and lowering the forks. The device shall be capable of measuring a minimum weight of 100 pounds (45 kg)and a maximum weight equal to the rated capacity of the truck plus 300 pounds (136 kg). Weighing device shall be capable of being adjusted for 25% tare. Either a digital readout or a bourdon tube dial indicator shall be provided. The entire instrument and connections shall be moisture and weather resistant. The weighing device shall be capable of withstanding the pressure caused by placing 150-percent of truck rated load on the forks without sustaining damage. The readout shall be readable by the operator when in the seated position. The weighing device shall read to an accuracy of 5 per cent of full scale reading as demonstrated when tested in accordance with 4.3.2.24.

3.3.5.7.1 Dial indicator. The dial shall be not less than 12 inches (300 mm) in diameter, and the increments shall be marked in 50-pound (23 kg) increments on gauges of 10,000-pound (4535 kg) or lesser capacity and in 100-pound (45 kg) increments on gauges of greater than 10,000-pound (4535 kg) capacity. The dial shall be capable of being rotated 360 degrees by means of an external tare adjusting knob. Bearing points or locking devices shall prevent dial rotation due to shock vibration. The indicating pointer shall be capable of 360 degrees rotation for full scale reading.

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The entire indicator case shall be filled with a liquid to provide movement lubrication and dampening of vibration-caused tube and pointer oscillation. There shall be sufficient air space left to allow for oil expansion owing to temperature changes. The indicator case shall be appropriately sealed to prevent leakage. The indicator shall be fitted with a fully adjustable pulsation damper located in an accessible position on the case.

3.3.5.7.2 Digital indicator. The digital indicator minimum increments shall be 50 pounds (23 kg) for indicators up to 10,000 pound (4535 kg) capacity and 100 pound (454 kg) increments on indicators greater than 10,000 pound (4535 kg) capacity. The digits shall have a minimum height of .75 inches (20 mm) visible to direct sunlight and subdued night lighting. The indicator shall have an internal zero and span adjustment for initial calibration and taring out the weight of the forks. The indicator shall have an external push-button auto zero device to tare out extraneous weights on the fork. Indicator must maintain auto zero reference when power is turned on or off or and indicate loss of auto zero reference. Indicator must indicate a truck overload condition of tare load and live load on the forks. The zero button must have a locking device to prevent actuation caused by vibrations.

3.4 Controls and instrumentation. All controls shall be provided with position markings (decals are not acceptable) either on a diagram visible to the operator or embossed or depressed on or near the control. Markings for controls which control the lift mechanism shall conform to ASME/ANSI B56.11.3. Letter height for control markings shall be equal to or greater than that specified for general dial and panel design for the applicable viewing distance of the 95th percentile man, with seat adjusted to rear position, as specified in MIL-STD-1472. Unless otherwise specified herein, the distance from the near edge of one control to the near edge of an adjacent control or other surface shall be a minimum of 1.5 inches (38 mm). Electric controls and instrumentation shall be moisture and weather resistant. Mechanical push-pull remote controls when furnished shall be of the waterproof antifriction type, conforming to requirements for marine cables in SAE J917, except as specified herein. As a minimum, all exposed parts of the control cable shall be fabricated of corrosion resistant steel. The cables shall be covered with a polyethylene or vinyl jacket. Cable ends shall be tailored to actuate all levers through their respective arcs without binding. Bend radii shall be as recommended by the cable manufacturer plus an increase in radii of 20 percent. The as-installed mechanical efficiency of each installation shall be not less than 71 percent, and the no-load friction shall not exceed 5 ounces per foot (4.6 N/m). The manufacturer's published ratings for continuous operation shall not be exceeded for any required operation of the control specified herein. Forces necessary to actuate directional and hydraulic controls shall be 10 pounds (45 N) maximum. All hand-control levers and linkages shall withstand a minimum force of 50 pounds (225 N) applied at the handgrip in the normal directions of operation and shall have sufficient strength to withstand, without permanent deformation or damage, all vibration forces generated during all operations specified herein. The travel of the controls shall not exceed 6 inches (150 mm) from the center position when measured at the end of the grip.

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3.4.1 Light switch. A switch for simultaneous operation of the flood light(s) (see 3.5.1 if specified) and tail lamp shall be furnished in a location convenient to the operator.

3.4.2 Brake pedal. The brake pedal shall be located for right foot operation within a clear space at least 6 inches (150 mm) wide. Pedal width shall be 3 inches (76 mm) minimum. Grease fittings or permanent lubrication shall be provided for lubricating moving parts of the brake pedal and shaft.

3.4.3 Accelerator control. The accelerator control shall be installed for right-foot operation; it shall be located to the right of and shall be free of interference from the brake pedal by a distance of not less than 1.5 inches (38 mm).

3.4.4 Directional controls. Selective forward and reverse control shall be furnished. The control shall be located for foot operation or left hand operation. Movement of the hand control lever in the forward or upward direction shall cause forward motion of the truck.

3.4.5 Lift and tilt controls. Lift and tilt controls shall be self-centering lever(s), located for right-hand operation. The tilt control shall be at the right of the lift control when 2 levers are employed. Lift and tilt controls may be combined in a single control lever. Rearward or upward motion of lift and tilt control lever(s) shall raise or tilt load rearward.

3.4.6 5_. When sideshift attachment is furnished, the controls shall be self-centering lever(s), conveniently located to the right of the tilt control for right-hand operation. Rearward movement of the sideshifter control shall shift the forks to the right and forward motion of the lever shall shift the forks to the left. The attachment and its components shall show no evidence of deformation when tested as specified in 4.3.2.21.

3.4.7 Master switch. A master switch shall be furnished that has provision for a key. When key is removed, control circuits shall be opened or be interrupted making it impossible for any electrical control circuit to be activated. All trucks under each contract shall be keyed identically. Unless otherwise specified (see 6.2), two (2) keys shall be supplied with each truck. All keys shall be identical for every truck under each contract.

3.4.8 Instruments. All gages, indicator lights, and meters shall be flush mounted on the instrument panel in view of the operator when he is in operating position. All instruments and exposed electrical components shall be weather resistant.

3.4.8.1 Hour meter. An electrically operated hour meter which registers the total number of hours of operation of the traction and pump motors shall be provided. Hour meters for E and EE trucks shall be in accordance with MIL-M-3971, type I, Grade A. Hour meters for EX trucks shall meet ANSI/UL 583 requirements. Meter shall be weather resistant.

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3.5 Accessories. When applicable the provisions of UL 583 shall apply. Electric accessories and instrumentation shall be moisture and weather resistant when tested in accordance with Section 4.

3.5.1 Flood lights. Each truck, except type EX truck, shall be equipped with one sealed beam, 25 watt minimum, floodlamp, shock mounted in an elastometer ring housing on the left side of the mast. Directional adjustment range of lights shall be a minimum of 45 degrees above and below the horizontal plane. Horizontal adjustment range shall be a minimum of 300 degrees to illuminate front and rear areas. When specified (see 6.2) trucks shall be equipped with an identical floodlight mounted on the right side of the mast. When specified (see 6.2) one rear facing flood light shall be mounted on the rear portion of the truck. Adequate protection against damage, by means of position or guard, shall be provided. Type EX trucks shall have no flood lights. Each floodlight shall be furnished with three set of lenses. Color of lenses shall be clear, red and yellow. Red and yellow lenses shall be stored in a pocket located on the back of the operator's seat.

3.5.2 Taillight. One automotive, red, reflector-type, combination stoplight and tail lamp, of not less than 15 candle power (candela) and 3 candle power (candela) respectively, shall be recessed or mounted on the rear end within the plan outline of the truck and shall be so located that it will not interfere with the removing or installing of the battery. Bulbs shall be replaceable without removal of battery or counterweight. The stoplight shall operate automatically only upon actuation of the service brake pedal and shall not operate when parking brake is applied. The brake lights shall override hazard/emergency flashers. Type EX trucks shall have no stoplights or taillights.

3.5.3 Horn. Truck shall be equipped with an electric horn (or air horn for EX truck) with the activating switch located within easy reach of the operator while seated on the truck. Horn button assembly and electrical wiring for the horn shall be constructed to be moisture and weather resistant.

3.5.4 Driver's overhead guard. An overhead safety guard in accordance with ASME/ANSI B56.1 shall be furnished. When specified (see 6.2), trucks shall be equipped with a high strength overhead guard capable of withstanding the tests specified in 4.3.2.10. Where the upright in the collapsed position is lower than the overhead guard height, the guard shall be removable with hand tools and no component shall be mounted on the overhead guard. A minimum of 39 inches (990 mm) from top of seat to underside of guard shall be provided at all tilt positions of the mast. Effective dimension measurement point on the seat shall be the lowest point on the seat cushion surface when the seat is occupied by a 150 pound (68 kg) operator seated in driving position. Guard height, with the mast tilted rearward, measured from the ground level to the top side of guard shall not exceed the height on the appropriate specification sheet. Guard design shall not interfere with operation of the truck, nor with normal movements of the operator when entering, leaving, or operating the truck. Guard shall not interfere with servicing of the battery and shall permit removal of the battery, if from the top, in one continuous lift and swing operation. Overhead guard shall remain firmly secured to its support brackets under normal and test course operation.

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3.5.5 Fire extinguisher. When specified (see 6.2), each truck shall be equipped with a stored pressure type, potassium bicarbonate fire extinguisher. Mounting shall be on a positive locking quick release bracket, in an accessible position, with the extinguisher nozzle protected against damage.

3.5.6 Battery protector. A battery discharge indicator shall be provided. The indicator shall be a solid state device requiring no adjustment except a periodic light bulb check or replacement. It shall be tamper proof and carry UL approval. It shall be designed to protect the battery from approaching a dangerously low state of discharge by first warning the operator by means of a signal light, or electrical meter and signal light. After a short interval of time or after the consumption of an additional amount of electrical energy, it shall interrupt the lift control circuit making the lift mechanism inoperative until the battery has been replaced or recharged. The circuit shall be such that the operator cannot activate the lift control circuit by unplugging and replugging the battery after it has been 80 per cent discharged, (typically 1150 specific gravity). Unplugging and replugging the battery prior to reaching 80 per cent discharge shall not reactivate the lift control circuit. The discharge indicator shall reset when a battery charged to 56 per cent of full capacity (typically 1200 specific gravity) is put on the truck. The discharge indicator may have controls to adjust the discharge setting and the reset level to values other than 80 per cent and 56 per cent, provided such settings can be made without electrical test equipment and the positions of the settings are indicated by indelible markings on the controls. When an electrical meter is used, it shall be mounted on the driver panel, and it shall display energy in the method of empty to full type display used in gasoline vehicles. There shall be a defined point on the meter where the warning light goes on and a defined point where the truck lift mechanism becomes inoperative. Location and method of mounting shall provide plain visibility of the indicator to the driver without creating interference with truck operation, maintenance, and entry and egress movements of the operator.

3.6 Performance.

3.6.1 Lifting speed. Minimum lifting speed over the entire distance from ground level to maximum fork height with rated load (ambient temperature between 65°F (18°C) and 85°F (29oC)) shall be as specified on the appropriate specification sheet, when tested in accordance with 4.3.2.2.

3.6.2 Lowering speed. Minimum speed of lowering of unloaded forks (ambient temperature between 65°F (18°C) and 85°F (29°C)) shall be as specified in the appropriate specification sheet, over the entire distance from maximum fork height to ground level when tested in accordance with 4.3.2.2. Speed of lowering forks with rated load shall be not more than 80 feet (24 m) per minute over the entire distance from maximum fork height to ground level. The hydraulic control system shall be damped, metered, or easily manually regulated to reduce shock and prevent the truck from overturning when a rated load is lowered at maximum speed and then stopped quickly.

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3.6.3 Right-angle turn. Trucks, when carrying rated load, shall be capable of backing through a turn in either direction and returning to the same or a parallel position within dimension "A", as measured by 4.3.2.3. Dimension "A" shall be as specified on the appropriate specification sheet.

3.6.4 Steering. The steer wheels shall be capable of being turned from extreme right to extreme left or vice versa in not more than 6 complete turns of the hand steering wheel, and the force required to operate within the first half of the inner steer angle (from a straight-ahead position) shall not exceed 40 pounds (178 N), nor shall it exceed 150 pounds (670 N) at the extreme ends of the steer in either direction when the unloaded truck is stationary on a smooth, dry concrete floor as measured in 4.3.2.9. When power assist or full power is applied, the maximum steering effort through the entire turn shall be not less than 2 pounds (9 N) nor more than 25 pounds (110 N) when measured under the same conditions. Wear of the steering mechanism shall be measured in accordance with 4.3.2.9. Free play shall not exceed 3 inches (76 mm) for manual steering and power steering with the pump motor running. The steering gear assembly shall be capable of withstanding an unbalanced tangential force applied at the rim of the hand steering wheel without permanently distorting or deforming any part of the steering mechanism as tested in 4.3.2.9. There shall be a minimum of 7 inches (180 mm) measured vertically between the lowest point on the steering wheel rim and a line parallel to the floor passing through the highest point of the unoccupied seat cushion when seat is held in full down position. There shall be a minimum of 3 inches (76 mm) between rim of steering wheel and directional control lever. The truck shall negotiate the obstacle course as specified in 4.3.2.35 with only the displaced wheel off the ground at any time.

3.6.5 Speed. While carrying rated capacity on a level surface, trucks shall be capable of attaining specified speed listed on applicable specification sheet in forward and reverse directions. Speed shall be as measured in accordance with 4.3.2.4.

3.6.6 Slope ascension, forward direction. Trucks with and without rated load shall be capable of ascending the grade specified on the applicable specification sheet. Test shall be conducted on a dry concrete, asphalt or steel surface. Truck shall be able to accelerate from a dead stop on this slope, when carrying a load. Shipboard trucks (see 6.2), shall ascend grade in forward direction only when loaded. Conformance to these requirements shall be tested as specified in 4.3.2.5.

3.6.7 Upright tilt. When loaded, truck masts shall have forward and rearward tilt not less than that specified on the applicable specification sheet. Positive means shall be furnished to prevent cavitation of the tilt cylinders when tilting rated load to full forward tilt with oil temperature not less than 120°F (49°C). Tilt shall be measured from a true vertical when tested in accordance with 4.3.2.6.

3.6.8 Collapsed mast height. Maximum collapsed mast height of unloaded truck measured from the ground to the top of mast of the fixed uprights in a vertical position shall be as specified on the appropriate specification sheet when measured in accordance with 4.3.2.1.

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3.6.9 Minimum fork height. The minimum fork height of a truck bearing rated capacity load and the uprights fully extended vertically, measured from the ground to the load carrying surface of the fork at the rated load center shall be not less than that specified on the applicable specification sheet when measured in accordance with 4.3.2.1.

3.6.10 Free lift height. Lift of forks without increase in the specified collapsed mast height shall be not less than that specified in the applicable specification sheet. Free lift height shall be measured in accordance with 4.3.2.1.

3.6.11 Underclearance. The truck with rated load shall have sufficient underclearance to permit safe operation from one horizontal plane to another, using the slope specified by 3.6.6. Underclearance beneath the mast assembly, when in a true vertical position and subjected to rated load, shall be not less than 2.5 inches (63 mm). Hydraulic fittings, hoses, tubing, and linkages shall not be the lowest portion of the truck and shall be protected by structural members from striking obstacle.

3.6.12 Lift assembly drift. The lift assembly shall be capable of holding the rated load at maximum height not less than 2 minutes with not more than 1.75 inch (44 mm) of vertical drift and not more than 1 degree of rotational drift from the vertical when tested in accordance with 4.3.2.7. Specified drift rates shall be measured when hydraulic oil temperature at the inlet to each cylinder is not less than 120°F(49°C).

3.6.13 Forks and mast assembly. Forks, mast assembly, and related structures shall show no permanent deformation or failure when subjected to test of 300 percent of rated load as specified in 4.3.2.12. For shipboard trucks (see 6.2) first produced truck shall be subjected to 300 per cent of rated load and quality conformance test shall be at 150 per cent of rated load.

3.6.14. Hydraulic system. The hydraulic system shall show no leaks or breakage when the truck is subjected to the test prescribed in 3.6.13 with the pressure relief valve inoperative. This requirement does not include the pump. The pressure relief valve shall open when pressure equals no more than 150 percent of the maximum pressure needed to lift rated load to the maximum fork height.

3.6.15 Service brakes. The brake system shall be capable of withstanding brake pedal force of 250 pounds (1110 N) without failure of any component. This requirements shall be tested according to 4.3.2.15. Brake parts designed to be rigid shall not loosen as a result of this test.

3.6.16 Stability. Except for shipboard trucks, the truck shall conform to the stability requirements of ASME/ ANSI B56.1 when tested as specified in 4.3.2.8. Trucks for shipboard and shorebased ordnance handling use shall meet the stability requirements of the applicable specification sheets when tested as specified in 4.3.2.8.

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3.6.16.1 Shipboard longitudinal stability.

3.6.16.1.1 Stacking. In accordance with the procedure specified in Test Method No. 8, the shipboard truck carrying rated load at the specified maximum elevation, shall not overturn when the platform upon which the truck is placed is tilted to the specified slope.

3.6.16.1.2 Traveling. In accordance with the procedure specified in Test Method No. 8, the shipboard truck, carrying rated load at 12 inches (305 mm) above the platform on which the truck is placed, shall not overturn when the platform is tilted to the specified slope.

3.6.16.2 Shipboard lateral stability.

3.6.16.2.1 Stacking. In accordance with the procedure specified in Test Method No. 8, the shipboard truck, carrying the rated load at the specified maximum elevation, shall not overturn when the platform on which the truck is standing is tilted to the specified slope.

3.6.16.2.2 Traveling. In accordance with the procedure specified in Test Method No. 8, the shipboard truck, without load, shall not overturn when the platform on which the truck is standing is tilted to the specified slope.

3.6.17 Overhead guard strength. The overhead safety guard shall be capable of withstanding the impact of a 100-pound (45 kg), solid hardwood cube (or equivalent) dropped a distance of 5 feet (1.5 m), 10 times, without fracture and without permanent deformation exceeding 0.75 inch (19 mm). It shall be of sufficient strength to support a uniformly distributed static test load equal to the applicable loads as specified in ASME/ANSI B56.1 and the appropriate specification sheet when tested according to 4.3.2.10. When a high strength overhead safety guard is specified (see 6.2), the guard shall be capable of withstanding an 8-foot (2.5 m) drop of a 500-pound (225 kg) high density cube, 10 times, without failure or permanent deformation exceeding 3 inches (76 mm) when tested according to 4.3.2.10.

3.6.18 Stopping distance. Truck, with rated load, shall be capable of stopping from a rearward maximum speed in accordance with the applicable requirements of ASME/ANSI B56.1 when tested as specified in 4.3.2.13 except brake pedal force should not be greater than 110 pounds(485N). Brake parts designed to be rigid shall not loosen as a result of this test.

3.6.19 Mechanical parking brake. The parking brake when activated by the seat shall be capable of retaining the truck with rated load on a slope equal to the specified slope ascension in both forward and reverse direction. If truck is specified for shipboard use (see 6.2), test of parking brake is to be accomplished only with the truck ascending specified slope in forward direction. When the truck is equipped with a hand-operated brake, it shall be capable of retaining the truck on a grade equal to the specified slope ascension under the same condition. Tests shall be in accordance with 4.3.2.5.

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3.6.20 Resistance to saline atmosphere. When resistance to saline atmosphere is specified or when truck is specified for shipboard use (see 6.2), all components shall be designed and installed to withstand the corrosive effect of the salt fog test specified in 4.3.2.20 without any loss of performance and without sustaining harmful corrosion.

3.6.21 Power consumption. Unless otherwise specified (see 6.2), the truck with battery shall meet the power consumption requirements of 4.3.2.23.

3.6.22 HI (High-Impact) shock requirement. When specified (see 6.2), the truck and the battery as a component of the truck shall be designed to withstand Type A HI Shock as a grade A item in accordance with MIL-S-901 when tested in accordance with 4.3.2.25. See paragraph 6.3 for reporting requirement.

3.6.23 Rain. The truck shall be capable of performing all its normal operations after being subjected to the rain test specified in 4.3.2.26.

3.6.24 Truck weight. When specified on the applicable specification sheet, the truck shall not exceed the weight listed when tested in accordance with 4.3.2.22.

3.6.25 Reliability. Truck shall complete 2000 laps without failure when tested as specified in 4.3.2.35. Failure shall be as defined in 4.3.2.35. If failure occurs before lap 2000, the cause of failure shall be corrected and the test restarted at the first lap.

3.7 Workmanship. The truck shall withstand any operation specified herein without permanent deformation, breakage of connections, malfunction, or component interference caused by incorrect workmanship. All parts of the truck before and after painting shall be clean and free from sand, rust, dirt, fins, pits, spurs, scale, flux, and other harmful extraneous material. Edges and surfaces exposed to operating and maintenance personnel shall be smooth and rounded to the extent that a hazardous surface does not exist.

3.7.1 Riveted connections. Rivets shall fill the holes completely. The upset rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member, and shall be in accordance with SAE J492.

3.7.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subject to proof and service loading. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the forklift to conform to the examination and test requirements specified in Section 4. Parts to be joined by filler welds shall be brought into as close contact as possible and in no event shall be separated by more than 0.187 inch (5 mm) unless appropriate bridging techniques are used. The welding process used in fabrication of the forklift shall be at the option of the supplier.

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3.7.3 Bolted connections. Bolt holes shall be accurately formed and shall have the burrs removed. Washers, lockwashers or other positive locking devices shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement.

3.7.4 Steel and other metal fabrication. Metals used in the fabrication of the truck shall provide original quality surface finish and shall be free from kinks and sharp bends. Metals having corroded or pitted surfaces are not acceptable. The straightening of materials shall be done by methods that will not cause weakening or injury to the material. Burrs and sharp edges in holes and on sheets, plates, and members shall be removed sufficiently to assure correct fits and to prevent loosening of fasteners and damage to components. from re-entrant cuts. Heated metals shall be allowed to cool slowly, except for designed heat treatment, and over-heating shall be avoided in accordance with the recommendations of the metal manufacturer. All modular assembly fabrication shall provide for interchangeability of components.

3.7.5 Machine work. Tolerances and gauges for metal fits shall conform to the limits specified herein.

3.7.6 Castings and forgings. All parts, components, and assemblies of the forklift which include castings and forgings shall be clean of extraneous material such as sand, dirt, sprues, scale and flux. Rework shall be limited to procedures which do not reduce mechanical properties or affect function. Castings with fastener holes shall provide for correct seating of fasteners such as bolt, lockwasher and nut assemblies and capscrews.

3.8 Treatment and painting. All exterior surfaces of the truck shall be thoroughly cleaned and shall be dry and free from mill scale, oil, grease, dirt and rust and shall be painted as soon as practicable after cleaning with not less than one coat of primer and two coats of synthetic enamel. Primer and finish coats shall be applied to assure complete coverage and durability of the finish. The finish color for the type E and EE trucks shall be color 13538 of FED-STD-595. The finish color for the type EX truck shall also be 13538, together with a single 6-inch (150 mm) wide blue diagonal stripe at 45 degrees across the right, left, and rear of truck; in addition, the type EX truck shall be identified as to type by painting 4-inch (100 mm) high black block letters with "EX" in a conspicuous space at right, left, and rear of truck. The overhead guard and mast may be the manufacturer's standard matte black color. When a first produced truck is required for test and evaluation, it shall be painted with one primer coat only. The finish coat, when dry, shall be a smooth even surface, free from runs, sags, peels, chips, blisters, areas of thin film or no film. Surfaces of components and assemblies of the truck not normally painted with a finish coat shall be cleaned and prepared in accordance with good commercial practice. Lift chains, sprockets, rollers, plastics, fabric hoses (except hydraulic hoses) and drive belts shall not be painted except that slight overspray will be permitted provided there is no interference with the proper functioning of the component. Paint must be kept off working surfaces where interference with working of parts would result.

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3.8.1 Walkway coating. Floor plates and step surfaces shall be painted with type IV , color number 39038, black, of MIL-W-5044 and FED-STD-595, or may be covered with the manufacturer's standard oil and grease resistant rubber matting.

3.9 Electromagnetic interference control. When specified (see 6.2), shipboard trucks shall be equipped for control of electromagnetic interference in accordance with Part 8, Class C1, Group II of MIL-STD-461. Shorebased trucks shall be equipped for control of electromagnetic interference in accordance with SAE J551C. Tests shall be in accordance with 4.3.2.19.

3.10 Identification marking. Each individual forklift and each separable attachment removable for either surface or air transportable purposes (e.g., cab, ROPS, fork carriage or counterweights) except forks shall be identified in acordance with MIL-STD-130 on identification plates conforming to MIL-P-514, type I, style 1, composition C (GG-P-455, type I, grade A, class 1) material. Additional or specified information to be indented or embossed on the plate(s) shall be as specified (see 6.2). Plates shall be furnished and installed by the contractor. A shipping data plate shall be furnished and installed which shall conform to MIL-P-514, type III, composition C (GG-P-455, type I, grade A, class 1) and in addition, shall indicate the silhouette of the forklift in transport position indicating the center of gravity, the location and capacity of the lifting and the tiedown attachments. The plates shall be attached by screws, drive screws, bolts, or rivets in conspicuous protected locations.

3.10.1 Instruction plates. The forklift shall be equipped with instruction plates conforming to MIL-P-514, type III, composition C (GG-P-455, type I, grade A, class 1) material, or commercially proven decals, including warnings, cautions and diagrams describing any special or important procedures to be followed in assembling, operating or servicing the forklift.

3.10.2 Wheel loading plate. Each forklift shall be equipped with a wheel loading plate conforming to MIL-P-514, Type III, Composition C, (GG-P-455 Type 1, grade A, class 1) material. Plate shall be attached by screws, bolts, or rivets in a conspicuous location. As a minimum a plate shall have the information shown in Table I.

TABLE I WHEEL LOADING PLATE

Wheel loading (no load on forks)	
Drive wheels (each side)	pounds(kilograms)
Steer wheels (each)	pounds(kilograms)
Wheel loading (rated load on forks)	
Drive wheels (each side)	pounds(kilograms)
Steer wheels (each)	pounds(kilograms)

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3.10.3 Body marking. The U. S. Military Property Number shall be indelibly stamped into the external plate of the truck body. The marking shall be located on the right side of the truck, approximately 6 inches (152 mm) below the center of the entrance to the operator's compartment. Marking shall be in 0.25-inch (6 mm) high block letters and numbers.

3.10.4 Safety Rating Each truck shall be equipped with a safety designation plate conforming to MIL-P-514, type III, composition C (GG-P-455, type I, grade A, class 1). As a minimum, the plate shall indicate the truck manufacturer's name, truck's complete model number, safety designation (Type E, EE, or EX), independent testing laboratory's name, independent laboratory's registration or index number assigned to the inspected truck, and the date and the location of independent laboratory's inspection.

3.10.5 Battery Plate Each truck shall be equipped with a battery information plate attached to the truck conforming to MIL-P-514, type III, composition C (GG-P-455, type I, grade A, class 1) material. As a minimum, the plate shall indicate battery weight, dimensions, ampere-hour capacity at a 6-hour rate, voltage, number of cells, number of plates, cable lead length and position.

3.11 Vehicle marking. The lifting capacity of the truck shall be painted in block numbers 3 inches (80 mm) inches high on each side of the mast. When space does not allow 3 inch (80 mm) numbers, 1 inch (25 mm) numbers may be used on the mast. The assigned registration number for each truck shall be painted in block numbers three 3 inches (80mm) high on each side of the truck. For pneumatic tired trucks, the tire pressures for each tire shall be stenciled in block numbers 1 inch (25mm) high on each side of the truck near the applicable tire. Durable black enamel markers indicating the safety designation of the truck shall be applied on each side and on the rear of the truck in a visible but protected location. These markers shall be in accordance with ANSI/UL 583. In addition, supplementary marking and load handling symbols as specified in ASME/ANSI B56.11.3 are also required. Slinging and tiedown markings shall be in accordance with MIL-STD-209. "No rider" warning shall be painted in block numbers 2 inches (25mm) high on the rear of the mast. Block numbers shall be painted in black enamel.

3.11.1 Special marking. When specified (see 6.2), trucks shall be marked in black color number 17038 in accordance with FED-STD-595. When the mast is painted black, markings placed on the mast shall be in white enamel. Marking paint shall be in accordance with TT-E-489. Characters shall be block-type capitals and Arabic numerals. Marking location and information and character size shall be as shown in Table II.

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TABLE II SPECIAL MARKING

Information	Example	Location	Character Size
Capacity	10,000 Lbs. (4540 kg)	On each side of mast	2 inch (50 mm) height
Rider warning	NO RIDERS	On rear of mast	2 inch (50 mm) height
Service designation and U S military property number	USN 13-XXXXX	On each side and rear	1-1/2 inch (38 mm) height

3.11.2 Cold Storage Marking. Trucks with cold storage protection features shall be marked with a single 6-inch (150 mm) wide diagonal side stripe, white, (Color Number 17886 of FED-STD-595). The stripe shall bisect each side of the vehicle at a 45-degree angle from lower rear of truck. In addition, trucks for cold storage use shall be identified with words COLD STORAGE USE APPROVED in black (Color Number 17038 of FED-STD-595) 1.5 inch (38 mm) high block type capital characters located in a conspicuous space on each side of the truck.

3.11.3 Shipboard Marking. Shipboard trucks (see 6.2) shall be identified with words SHIPBOARD USE APPROVED in black (Color Number 17038 of FED STD-595) 1 1/2 inches (38mm) high block type capital chartacters located on each side and rear of the truck.

3.11.4 Structural testing marking. Shipboard trucks (see 6.2) shall be identified with the words:STRUCTURALLY TESTED, the test date and the manufacturer's name in black (color Number 17038 of FED-STD-595) 1.5 in (38mm) high block type capital characters located on each side of the truck.

3.12 Fungus and moisture resistance. When specified (see 6.2), the electrical circuitry, including all components and connections except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, Composition I or II as applicable (see 3.12.1), with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish:

(a) Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.

(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush or a combination of both to give a minimum dry-film thickness of 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

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3.12.1 Composition. Composition II shall be used only in the case when local air pollution regulations for the application of the varnish precludes the use of Composition I. When Composition II is used, the supplier shall provide evidence to the Government that the use of Composition II is required and shall certify that the Composition II material complies with Rule 66, Air Pollution Control District, County of Los Angeles, CA.

3.13 Transportability. To the extent specified below, the forklift trucks shall be transportable worldwide by the highway, rail and water modes without special permits. Transportability criteria are defined in MIL-STD-1366 and MIL-HDBK-157. Vehicles, trailers and all other materiel shall not be loaded beyond their design ratng at any time.

3.13.1 Highway. When specified (see 6.2), the forkift trucks shall not exceed legal size and weight limits for highway transport imposed by the states and North American Treaty Organization (NATO) countries. United States and NATO statutory road limitations are available in the American Trucking Association (ATA) Inc.,Summary of Size and Weight Limits and the International Road Federation (IFR), Limits of Motor Vehicle Sizes and Weights respectively.

3.13.2 Rail. When specified (see 6.2), the forklift trucks, when loaded on a 50 inch (1270 mm) high railcar, shall meet the dimensional requirements of the Association of American Railroads (AAR) Outline Diagram for Single Loads, Without End Overhang, on Open-Top Cars and the Gabarit International de Chargement (GIC) gauge shown in STANAG 2832, which apply to standard-gauge rail lines in the Continental United States (CONUS) and European countries. When specified (see 6.2) , the forklift trucks shall withstand the rail impact test specified in MIL-STD-810 without degradation or damage.

3.13.3 Water. When specified (see 6.2), the forklift trucks shall be transportable on breakbulk (general cargo), roll on/roll off (RORO), and barge-carrying (LASH and SEABEE) ships witout disassembly. The forklift trucks shall be transportable on the LACV-30, LARC-LX, LCM-8, LCU 1646 or LCU 2000 and larger vessels/ships.

3.13.4 Air. When specified (see 6.2), the forklift shall be air transportable in the C-130 and C-141 aircraft without removal of any component and by driving on and off the aircraft from and to ground level. The truck shall meet or exceed all requirememts of MIL-STD-1791 and AFSC Design Handbook DH1-11.

3.13.5 Certification. Acceptable evidence of meeting transportability requirements specified is a letter from the Military Traffic Management Command, Transportation Engineering Agency, 12388 Warwick Blvd, P.O. Box 6276, Newport News, VA 23606-0276 certifying that drawings for trucks being furnished have been examined and specified transportability requirements have been met.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) specified herein.

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Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet the requirements of section 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all the requirements of the contract. Sampling inspection, as a part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections and tests. The inspection requirements specified herein are classified as follows:

(a) First produced truck inspection and tests (see 4.3).

(b) Quality conformance inspection (see 4.4).

(c) Examinmation of preparation for delivery (see 4.5).

4.3 First produced truck inspection. Prior to examination and test of the truck specified in 3.1.1 the following shall be performed:

(a) Furnish a schedule of maintenance to be followed during all testing of the first produced truck. A list of special tools required to perform this maintenance shall be provided.

(b) Service the truck with oils and greases specified herein and designated for use in the ambient temperature at which the tests will be conducted.

(c) Break in the truck as prescribed by the supplier.

4.3.1 Inspections.

4.3.1.1 Pretest examination Prior to testing according to 4.3.2, one or more of the first produced trucks as specified in 3.1.1 shall be examined for the defects marked "X" in Column 1 of Table III.

4.3.1.2 Post-test inspection (see 4.3.2.38).

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4.3.2 Tests. Upon successful completion of the inspection specified in 4.3.1.1, the truck shall be subjected to the tests marked "X" in Column 1 of Table IV. Acceptance of a first produced truck shall not exclude the remaining trucks from the quality conformance inspection and acceptance provisions specified in Section 4. Test conditions and schedule shall be as follows:

(a) Unless otherwise specified herein, tests shall be conducted at the ambient temperature and climatic conditions existing at the place of test. The test surface, unless otherwise specified for specific tests, shall be level within +/- 1 degree. The surface of the test area shall produce a coefficient of friction of .5 or greater for trucks equipped with cushion tires, or .5 or greater for trucks equipped with solid tires. (If the specified coefficient of friction is not available, it is permissible to cover the travel surface with a friction increasing material). Only that maintenance established by the supplier and submitted as a maintenance schedule prior to commencement of tests shall be performed during the tests.

(b) Tests shall be in accordance with Table IV and may be conducted in any order desired except that reliability and post test shall be the last tests conducted. An "X" in the applicable column indicates the tests that shall be conducted.

(c) Rated load, test load, capacity load and calibrated load shall be established with a load equivalent to an unrestrained cube with overall dimensions twice the load center dimension and whose center of gravity is located at the geometric center of the cube. The tolerance for weight shall be plus or minus 1 per cent.

(d) Load center shall be defined as the horizontal longlitudinal distance from the intersection of the horizontal load carrying surfaces and vertical load engaging faces of the forks to the center of gravity of the load. The tolerance for dimensions shall be plus or minus 0.5 per cent.

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Table III. Examination Schedule

First Produced Inspection	Quality Conform	Defect	Requirement Paragraph
x		Material not as specified	3.1.2
x		Dissimilar metals not protected	3.1.2.1
x		Plating not as specified	3.1.2.2
x		Cold storage cond. not as spec.	3.1.2.3
x	x	Design not as specified not as specified	3.1.3
x		Safety not as specified	3.1.3.2
x		Bearings not as specified	3.1.3.3
x	x	Lubrication and fittings not as specified	3.1.3.4
x		Screw threads not as specified	3.1.3.5
x		Bumper strip not as specified	3.1.3.6
x	x	Maintainability not as specified	3.1.3.7
x	x	Special lubrication system not as specified	3.1.3.8
x		Resistors not as specified	3.1.4.1
x	x	Enclosures and shields not as specified	3.1.4.2
x	x	Overcurrent devices not as spec	3.1.4.3
x		Heat producing devices not as specified	3.1.4.4
x	x	Static discharge devices not as specified	3.1.4.5
x		Control panel not as specified	3.1.4.6
x		Conductors not as specified	3.1.4.7
x	x	Motors and motor connections not as specified	3.2.1.1& 3.2.1.2
x		Insulation not as specified	3.2.1.3
x		Speed controller not as spec	3.2.1.4
x		Safety cutout system not as spec	3.2.1.5
x	x	Battery not as specified	3.2.2
x		Battery connector mounting not as specified	3.2.3
x		Battery well not as specified	3.2.3.1
x		Drive unit not as specified	3.2.3.2
x		Hydraulic system not as specified	3.2.4
x	x	Chassis and frame not as specified	3.3.1
x		Uprights and carriage not as specified	3.3.1.1
x		Fork and fork carrier not as specified	3.3.1.2
x	x	Load backrest not as specified	3.3.1.3

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Table III. Examination Schedule (Con't)

First Produced Inspection	Quality Conform	Defect	Requirement Paragraph
x		Wheels not as specified	3.3.2
x		Tires not as specified	3.3.2
x		Steering not as specified	3.3.3
x		Service brakes not as specified	3.3.4.1
x		Parking brake not as specified	3.3.4.2
x		Truck body not as specified	3.3.5
x	x	Seat not as specified	3.3.5.1
x		Wheel guards not as specified	3.3.5.2
x	x	Towing hook not as specified	3.3.5.3
x		Slinging and tiedown attachments not as specified	3.3.5.4
x	x	Sideshift attachment not as spec	3.3.5.6
x	x	Weighing device not as specified	3.3.5.7
x	x	Controls and instrumentation not as specified	3.4
x	x	Floodlights not as specified	3.5.1
x	x	Taillight not as specified	3.5.2
x	x	Horn not as specified	3.5.3
x	x	Overhead guard not as specified	3.5.4
x	x	Fire extinguisher not as spec	3.5.5
x	x	Battery protector not as spec	3.5.6
x	x	Riveted connections not as spec	3.7.1
x	x	Welding not as specified	3.7.2
x	x	Bolt connections not as spec	3.7.3
x	x	Metal fabrication not as spec	3.7.4
x	x	Machine work not as specified	3.7.5
x	x	Casting and forgings not as spec	3.7.6
x	x	Treatment and painting not as as specified	3.8
x	x	Walkway coating not as spec	3.8.1
x	x	Identification marking not as as specified	3.10
x	x	Shipping data plate not as spec	3.10.1
x	x	Wheel loading plate not as spec	3.10.2
x	x	Body marking not as specified	3.10.3
x	x	Safety designation plate not as specified	3.10.4
x	x	Battery plate not as specified	3.10.5
x	x	Vehicle marking not as specified	3.11
x	x	Fungus and moisture resistance not as specified	3.12
x		Transportability not as spec	3.13

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Table IV. Test Schedule

First Prod.	Quality Conform.	Post Test	Test	Test Para.	Requir. Para.	Test Method No
x		x	Collapsed mast height, maximum fork height, free lift height	4.3.2.1	3.6.8 3.6.9 3.6.10	1
x	x	x	Lift speed, lowering speed	4.3.2.2	3.6.1 3.6.2	2
x			Right angle turn	4.3.2.3	3.6.3	3
x	x	x	Travel speed	4.3.2.4	3.6.5	4
x	x	x	Slope ascension, parking brake, underclearance	4.3.2.5	3.6.6 3.6.19 3.6.11	5
x		x	Tilt	4.3.2.6	3.6.7	6
x		x	Drift	4.3.2.7	3.6.12	7
x			Stability	4.3.2.8	3.6.16	8
x		x	Steering	4.3.2.9	3.6.4	9
x			Overhead guard	4.3.2.10	3.6.17	10
x			Lifting & tiedown provisions	4.3.2.11	3.3.5.4 & 3.3.5.5	11
x		x(*)	Mast Assy. Overload	4.3.2.12	3.6.13 & 3.6.14	12
x	x	x	Stopping distance	4.3.2.13	3.6.18	13
x			Towing hook	4.3.2.14	3.3.5.3	14
x	x	x	Brakes	4.3.2.15	3.6.15	14
x			Maintainability	4.3.2.16	3.1.3.7.1	15
x			Fork load backrest	4.3.2.17	3.3.1.3	16
x			Tire loading	4.3.2.18	3.3.2.2 3.3.2.4	17
x			Electromagnetic Interference characteristics	4.3.2.19	3.9	--
x			Saline atmosphere	4.3.2.20	3.6.20	--
x		x	Sideshift attachm't	4.3.2.21	3.4.6	--
x			Truck weight	4.3.2.22	3.6.24	--
x			Power Consumption	4.3.2.23	3.6.21	18
x	x		Weighing device	4.3.2.24	3.3.5.7	19
x			Hi-shock test	4.3.2.25	3.6.22	--
x			Rain protection	4.3.2.26	3.6.23	20
x			Frame racking	4.3.2.27	3.3.1	21
x			Operating temp.	4.3.2.28	3.1.3.1	22
x			Hydraulic pump vac.	4.3.2.29	3.2.4.1.4	--

(*) Quality conformance test for shipboard trucks only.

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Table IV. Test Schedule (Con't)

First Prod	Quality Conform	Post Test	Test	Test Para.	Requir. Para.	Test Method No
x			Suction line	4.3.2.29.1	3.2.4.1.4	—
x			Hydraulic pump pressure test	4.3.2.30	3.2.4.2	—
x			Hydraulic filter	4.3.2.31	3.2.4.3.1 &3.2.4.3.2	23
x		x	Hydraulic filter life	4.3.2.31.1	3.2.4.3.1 &3.2.4.3.2	—
x			Hydraulic cylinder	4.3.2.32	3.2.4.6	24
x			Hydraulic system Cleanliness	4.3.2.33	3.2.4.8	25
x			Hydraulic fittings	4.3.2.34	3.2.4.1.1	26
x			Reliability	4.3.2.35	3.6.25	27
x			Battery protector	4.3.2.36	3.5.6	28
x			Cold storage	4.3.2.37	3.1.2.3	29

4.3.2.1 Maximum fork height, collapsed mast height, free lift height. Test the truck in accordance with Test Method No. 1. Failure to conform with the requirements of 3.6.8, 3.6.9 and 3.6.10 shall constitute failure of this test.

4.3.2.2 Lift and lowering speeds. Test the truck in accordance with Test Method No. 2. Failure to conform with the requirements of 3.6.1 and 3.6.2 shall constitute failure of this test.

4.3.2.3 Right angle turn. Test the truck in accordance with Test Method No. 3. Failure to conform with the requirements of 3.6.3 and the individual specification sheet shall constitute failure of this test.

4.3.2.4 Speed. Test the truck in accordance with Test Method No. 4. Nonconformance to 3.6.5 and the individual specification sheet shall constitute failure of this test.

4.3.2.5 Slope ascension, parking brake, underclearance. Test the truck in accordance with Test Method No. 5. Nonconformance to 3.6.6, 3.6.19 and 3.6.11 shall constitute failure of this test.

4.3.2.6 Tilt. Test the truck in accordance with Test Method No. 6. Nonconformance to 3.6.7 shall constitute failure of this test.

4.3.2.7 Drift. Test the truck in accordance with Test Method No. 7. Nonconformance to 3.6.12 shall constitute failure of this test.

4.3.2.8 Stability. Test the truck in accordance with Test Method No. 8. Nonconformance with the requirements of 3.6.16 shall constitute failure of this test.

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4.3.2.9 Steering. Test the truck in accordance with Test Method No. 9. Nonconformance with the requirements of 3.6.4 shall constitute failure of this test. Other maneuverability features shall meet the endurance test requirements.

4.3.2.10 Overhead guard strength. Test the truck in accordance with Test Method No. 10. Nonconformance with the requirements of 3.6.17 shall constitute failure of this test.

4.3.2.11 Lifting and tiedown attachments. Test the truck in accordance with Test Method No. 11. Nonconformance to 3.3.5.4 and 3.3.5.5 and requirements of Test Method No. 11 shall constitute failure of test.

4.3.2.12 Mast assembly overload. Test the truck in accordance with Test Method No. 12. Nonconformance with the requirements of 3.6.13 and 3.6.14 shall constitute failure of this test.

4.3.2.13 Stopping distance. Test the truck in accordance with Test Method No. 13. Nonconformance with the requirements of 3.6.18 shall constitute failure of test.

4.3.2.14 Towing hook. Test the truck in accordance with the applicable part of Test Method No. 14. Nonconformance with the requirements of 3.3.5.3 shall constitute failure of this test.

4.3.2.15 Service brake pressure. Test the truck in accordance with Test Method No. 14. Nonconformance with the requirements of 3.6.15 shall constitute failure of this test.

4.3.2.16 Maintainability. Test the truck in accordance with Test Method No. 15. Failure to perform the operations in the allotted time as specified in 3.1.3.7.1 shall constitute failure of this test.

4.3.2.17 Fork load backrest. Test the backrest in accordance with Test Method No. 16. Nonconformance to 3.3.1.3 shall constitute failure of this test.

4.3.2.18 Tire loading. Test the truck in accordance with Test Method No. 17. Nonconformance with requirements of 3.3.2.2 or 3.3.2.4 shall constitute failure of this test.

4.3.2.19 Electromagnetic interference characteristics. The first produced truck equipped for electromagnetic compatibility in accordance with 3.9 shall be tested as specified in MIL-STD-461 or SAE J551C. (see 6.2) The supplier shall furnish the contracting officer the report of the test required by MIL-STD-461 or SAE J551C. Upon approval of the report by the contracting officer and provided other requirements of the specification are met, the approved truck shall be used as a model for all production trucks. Nonconformance to 3.9 shall constitute failure of this test.

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4.3.2.20 Saline atmosphere test. When specified for shipboard use or for resistance to corrosion in saline atmosphere (see 6.2), the truck shall be tested in accordance with MIL-STD-810, Test Method 509, Salt Fog. At the option of the supplier, the truck may be tested as a completely assembled unit or by components and subassemblies. If the truck is tested in the form of components and subassemblies, the following parts may be omitted: chassis, frame, body, counterweight, and battery furnished without cover. All other parts, including components, fasteners, subassemblies, battery furnished with cover, and the electric system shall be tested. Whether in assembled or in disassembled condition, the truck shall not be washed or operated in accordance with II-3 of Method 509 until the exposure and drying periods have been completed.

4.3.2.20.1 Assembled truck. Preparation of the assembled truck prior to exposure to salt fog shall be in accordance with II-2.5 of Method 509. All exterior surfaces of the truck, including the exposed surface of each piston rod and excluding only surfaces such as those of lift chains which are lubricated in service use, shall be cleaned and exposed to salt fog and the drying. The rod(s) of the tilt and of the sideshift cylinder(s), if provided, shall be at full extension. The lift cylinder rod which is exposed when the lift assembly is lowered in accordance with 4.3.2.20.3(c) herein shall be so cleaned and exposed. Components such as bearings which are lubricated in service use shall be lubricated in accordance with instructions of supplier prior to test.

4.3.2.20.2 Unassembled truck. Preparation of the truck tested as subassemblies and components shall be in accordance with II-2.5 of Method 509. Exterior surfaces of all of the parts shall be so cleaned, except those specifically omitted above and those which are lubricated in service use. Those surfaces, components, and subassemblies which are lubricated in service use shall be lubricated in accordance with instructions of the supplier prior to test. Surfaces to be cleaned and exposed to salt fog and for drying shall include the exposed surface of each piston rod. The rod(s) of the tilt and of the sideshift cylinder(s) if provided shall be at full extension for the cleaning and exposure. The lift cylinder rod surface which is exposed when the lift assembly is lowered in accordance with 4.3.2.20.3(c) shall be so cleaned and exposed.

4.3.2.20.3 Details. Pursuant to paragraph I-4.2 of Method 509, the following test information is given:

a. Exposure period shall be 72 hours followed by 168-hour drying period.

b. Configuration:

(1) Assembled truck resting on load tires with mast at full forward tilt and carriage at maximum sideshift to expose rod of shift cylinder if provided, and with mast lowered to where tips of forks rest on 4-inch (100 mm) x 4-inch (100 mm) wood blocks.

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(2) Unassembled truck parts supported as convenient with large pieces in positions similar to assembled positions described in (1) and resting on 4-inch (100 mm) x 4-inch (100 mm) wood blocks. Small pieces supported as nearly as possible to 30 degrees from vertical by cords, hooks, or racks.

c. Cyclic conditions not required.

d. Salt concentration 5 percent.

e. Additional guidelines:

(1) Visual inspection after 24-hour and 72-hour exposure, but no washing or operation until examination following full exposure and drying period. Truck shall be dried at room temperature while loosely covered with a sheet of clear plastic to retard drying and allow visual inspection.

(2) Evidence of harmful corrosion, loss of mobility of parts or inability to disassemble parts for service or repair shall constitute failure of this test. Operation of electrical system is required.

4.3.2.21 Sideshift attachment. When a sideshift attachment is specified as an accessory, the following test shall be performed. Operate the sideshift attachment for 100 full cycles at full speed from one extreme to the other with rated load on the forks. Test may be interrupted for 5 minutes every 25 cycles to allow system to cool. During the last 50 cycles, the control shall be released abruptly when the forks are at approximately mid-position between the left and right extremes, and then restarted. Upon completion of the 100 cycles, the attachment and its components shall be examined for compliance with 3.4.6. Any evidence of non-compliance shall be cause for rejection of the truck.

4.3.2.22 Truck weight. Truck shall be weighed to comply with 3.6.24. Any evidence of noncompliance shall be cause for rejection of the truck.

4.3.2.23 Power consumption. Test the truck in accordance with Test Method No. 18. Failure to conform with requirements of test method shall constitute failure of this test.

4.3.2.24 Weighing device. When a weighing device is specified it shall be tested according to Test Method No. 19 for conformance to the requirements of 3.3.5.7.

4.3.2.24.1 First produced truck. When a weighing device is furnished each truck, weighing device shall be adjusted to zero and marked in accordance with 4.3.2.24. By use of a load equal to the rated truck capacity plus or minus 1-percent, the accuracy of the weighing device to weigh the calibrated load shall be demonstrated. Failure to comply with this required accuracy shall be cause for rejection.

4.3.2.24.2 Production Truck. The weighing device on each production truck shall be tested according to paragraphs 1 through 3.f of Test Method No. 19."

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4.3.2.25 HI Shock test. When truck is specified to be designed to meet HI Shock requirement (see 6.2), a first produced truck with battery shall be tested by the contractor in accordance with MIL-S-901. The contractor shall locate a non-government shock test facility, arrange the test, and pay for the test. Prior to actual test, contractor shall submit a formal procedure to be used during the high shock test for approval. Procedure shall indicate applicable documents, test requirements, description of test, hardware configuration, preparation for testing, visual inspections, initial operational tests, evaluation of damage, shock test report, definition of failure, cause for rejection, and completion of shock testing/post shock examination. Pursuant to 6.1 of MIL-S-901 the following ordering data shall apply to the first article truck and battery which shall be mounted on the Floating Shock Platform during test and to the trucks furnished by the supplier:

(a) "Shock tests: HI (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements For", MIL-S-901.

(b) Issue of DODISS shall be the issue in effect on the date of invitation for bids or request for proposals.

(c) Grade A shock proofness required.

(d) The truck is class I, modified. The modification is that the truck rests on solid rubber or polyurethane tires which are similar in action to some resilient mountings.

(e) Type A test in which the truck is the principal unit and the battery is a subsidiary component.

(f) The truck is deck mounted.

(g) The truck is base mounted.

(h) The truck has unrestricted orientation.

(i) The truck with battery installed shall be secured to the deck simulator of the floating shock platform with tiedowns furnished by the testing laboratory.

(j) N/A

(k) The truck shall not be operated during the shocks, but it shall be operated briefly after each shock, to show lifting, lowering, drive wheels jacked up and rotating forward and reverse, sideshift operation if applicable steering.

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(l) Following the explosion of the depth charge at 20-foot (6 m) standoff, the truck and battery shall comply with (k) and the battery shall successfully demonstrate that it can reach rated capacity in accordance with 4.4.3 of W-B-133 except that not more than 2 cycles of charge shall be allowed after the HI shock testing. Minor permanent deformation of battery components and minor loss of electrolyte from splashing are permitted. Cracked or leaking cells, fractured intercell connectors, major loss of electrolyte, or failure to power the truck in accordance with (k) or to reach rated capacity in accordance with (k) shall constitute failure of the battery to meet the High Impact Shock requirement. Failure of the truck to comply with (k) or failure or permanent deformation of any component preventing operation of truck shall be regarded as failure of the truck to meet the H. I. Shock requirement.

(m) Acceptance authority is Navy Ships Parts Control Center, Code 0361, 5450 Carlisle Pike, P.O. Box 2020, Mechanicsburg, PA 17055-0788.

4.3.2.26 Rain protection. Test the truck in accordance with Test Method No. 20. Evidence of moisture in instruments or controls or inability to perform any normal function shall constitute failure of this test.

4.3.2.27 Frame racking. Test the truck in accordance with Test Method No. 21. Nonconformance with the requirements of 3.3.1 shall constitute failure of this test.

4.3.2.28 Operating temperature test. Test the truck in accordance with Test Method No. 22. Nonconformance with the requirements of 3.1.4.4 shall constitute failure of this test.

4.3.2.29 Hydraulic pump vacuum. Install a vacuum gage into the pump suction line immediately adjacent to the pump or at the pump. Install a thermocouple in the hydraulic reservoir. Run the pump at RPM encountered in truck operation. Measure the pump inlet pressure in inches of mercury vacuum at the specified temperature and speed. Nonconformance to 3.2.4.1.4 shall constitute failure of this test.

4.3.2.29.1 Suction line. The suction line(s) consisting of all fittings and hoses between the pump(s) and reservoir shall be tested by capping one end of the line and applying a minimum of 25 inches (84 KPa) of mercury vacuum to the other end of the line and holding the vacuum for a minimum of 1 minute. Coat all suction line junctures with detergent foam. Evidence of leakage or a reduction of outside circumference of more than 15 percent shall constitute failure of this test.

4.3.2.30 Hydraulic pump(s). Operate the pump(s) at maximum RPM encountered in truck operation and at 1.5 times maximum system working pressure for not less than 2 minutes. Check for external leaks at assembled surfaces and the shaft seal. Nonconformance to 3.2.4.2 shall constitute failure of this test.

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4.3.2.31 Hydraulic filter. Test the hydraulic filter in accordance with Test Method No. 23. Nonconformance to 3.2.4.3.1 or 3.2.4.3.2 as applicable shall constitute failure of this test.

4.3.2.31.1 Hydraulic filter life. The hydraulic filter life test shall be conducted on the truck during the reliability test. At conclusion of reliability test, establish maximum system oil flow through the filter at an oil temperature of 150°F (66°C) plus or minus 5°F (3°C), and note if element is bypassing. Nonconformance to 3.2.4.3.1 or 3.2.4.3.2 as evidenced by indicator reading bypass shall constitute failure of this test.

4.3.2.32 Hydraulic cylinders. Test each hydraulic cylinder in accordance with Test Method No. 24. Nonconformance to 3.2.4.6 shall constitute failure of this test.

4.3.2.33 Hydraulic system cleanliness. Test each truck in accordance with Test Method No. 25. Nonconformance to 3.2.4.8 shall constitute failure of this test.

4.3.2.34 Hydraulic fittings. Test each truck in accordance with Test Method No. 26. Nonconformance to 3.2.4.1.1 shall constitute failure of this test.

4.3.2.35 Reliability. The first-produced truck(s) shall be tested in accordance with Test Method No. 22. Testing shall be conducted at a rate of 20 to 25 laps/hr. Failure for the purpose of this test is defined as any malfunction which cannot be remedied by adjustment, repair, or replacement action by organizational maintenance using the organization tools and parts within 60 minutes and which may cause:

(a) Failure to commence operation, cessation of operation, or degradation of performance capabilities of the truck below designated levels.

(b) Serious damage to the truck by continued operation.

(c) Serious personnel safety hazard.

Simultaneous related malfunctions are considered as one failure.

4.3.2.36 Battery protector. Test the truck in accordance with Test Method No. 28. Nonconformance to 3.5.6 shall constitute failure of the test.

4.3.2.37 Cold storage protection. Test the truck in accordance with Test Method No. 29. Nonconformance to 3.1.2.3 shall constitute failure of this test.

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4.3.2.38 Post test inspection. Upon completion of tests in 4.3.2, the truck shall be subjected to the tests marked "X" in column 3 of Table II. Nonconformance with applicable requirements shall constitute a failure. Upon completion of all tests dismantle the truck sufficiently to permit detailed disassembly and visual inspection of the following components.

4.3.3 Inspection failure. Failure of a first-produced truck to meet any requirement specified herein, during and as a result of the examination and tests specified in 4.3.1 and 4.3.2 shall be cause for refusal by the Government to accept any production truck until evidence has been provided by the supplier that corrective action has been taken to eliminate deficiencies. Correction of such deficiencies shall be accomplished by the supplier at no cost to the Government on trucks produced under the contract. Any deficiencies found as a result of the first inspection and tests will be considered prima facie evidence that all trucks accepted prior to the completion of first inspection and tests are similarly deficient unless evidence to the contrary is furnished by the supplier and such evidence is acceptable to the contracting officer.

4.4 Quality conformance inspection.

4.4.1 Tests. Each truck shall be tested carrying a rated load as specified in Column 2 of Table II. Failure of any test shall be cause for rejection. Overheating, failure of any components, malfunction of any control, or evidence of leakage of fluids shall be cause for rejection of the truck. Unless otherwise specified (see 6.2), all tests shall be conducted utilizing a contractor furnished battery equal to the battery specified in 3.2.2.

4.4.2 Examination. After successful completion of all tests specified in 4.3.2, each truck shall be examined for the defects marked "X" in Column 2 of Table I. Presence of one or more defects shall be cause for rejection.

4.5 Examination of preparation for delivery. The preservation, packaging, packing, and marking shall be examined to determine compliance with MIL-STD-162.

5. PACKAGING

5.1 Preservation, packaging, and packing. Shall be in accordance with MIL-STD-162 or ASTM D3951 as specified (see 6.2).

5.2 Marking. Unless otherwise specified (see 6.2), marking for shipping and storage shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Intended use. Type E, EE, and EX trucks are intended for stacking, unstacking, moving general supplies, cargo and handling of propellants, ammunition, and other ordnance applications in and around warehouses, loading platforms, ordnance depots, ammunition depots, docks, and aboard ships. Trucks are intended for operation over paved and unpaved surfaces for short distances. Also, type EE and EX trucks are for use where continuous operation is required in slow speeds such as inching with minimum temperature rise of electrical components. Users are cautioned to use proper type of truck as prescribed by current safety regulations.

6.2 Acquisition requirement. Acquisition documents shall specify the following:

(a) Title, number and date of this specification.

(b) Issue of the DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

(c) Type and size required, applicable individual specification sheet. (see 3.1 and 3.2.1.1)

(d) When first article test is required (see 3.1.1)

(e) When truck is to be specified for shipboard use (see 3.1.2, 3.1.3.1, 3.1.3.4.1, 3.3.1.2, 3.3.2.2, 3.6.6, 3.6.16, 3.6.19, 3.6.20 and 3.11.3.

(f) When resistance to saline atmosphere is required for other than shipboard (see 3.1.2 and 3.6.20).

(g) When plating other than described is required (see 3.1.2.2).

(h) When special tools are not to be provided (see 3.1.3.7).

(i) When a special lubrication system is required (see 3.1.3.8).

(j) When battery is required to be furnished by contractor (see 3.2.2).

(k) When battery description is different from that specified (see 3.2.2).

(l) When battery connectors are different from that specified (see 3.2.2 and 3.2.3).

(m) When frame racking test is not required (see 3.3.1).

(n) When carriage load securing eyes are required (see 3.3.1.1).

(o) When other than hand adjustment of forks laterally (without hand tools) on trucks up to 10,000 pound (4540 kg) capacity is required (see 3.3.1.2)

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(p) When other than an auxiliary means of simultaneous adjustment of forks on 10,000 pound (4540 kg) capacity trucks is required (see 3.3.1.2).

(q) When forks shall be hydraulically adjustable from the operator's position for trucks of 10,000 pounds (4540 kg) capacity and above (see 3.3.1.2).

(r) When power steering is required (see 3.3.3).

(s) When parking brake with linkage actuating the service brake is required (see 3.3.4.2).

(t) When hand-operated parking brake lever with locking device is required (see 3.3.4.2).

(u) When slinging and tiedown provisions are required (see 3.3.5.4).

(v) When truck is required to be grade A item in type A High-Impact Shock (see 3.3.5.5,and 3.6.22).

(w) When sideshift attachment is required (see 3.3.5.6).

(x) When a weighing device is required (see 3.3.5.7).

(y) When number of keys other than specified is required (see 3.4.7).

(z) When additional floodlights are required (see.3.5.1).

(aa) When high strength overhead guard is required (see 3.5.4 and 3.6.17).

(ab) When fire extinguisher is required (see 3.5.5).

(ac) When power consumption test is not required (see 3.6.21).

(ad) When electromagnetic interference is required (see 3.9).

(ae) When classification of equipment for electromagnetic interference control shall be other than as specified (see 3.9).

(af) When vehicle marking other than as specified is required (see 3.11).

(ag) When vehicle requires special marking (see 3.11.1).

(ah) When fungus and moisture resistance is required (see 3.12).

(ai) Level and type of preservation, packaging and packing required (see 5.1).

(aj) When special highway, rail, water or air transportability is required (see 3.13.1, 3.13.2, 3.13.3 and 3.13.4).

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(ak) When rail impact test is required (see 3.13.2).

(al) Marking for shipment and storage, if different (see 5.2).

(am) When cold storage conditioning is required (see 3.1.2.3.)

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data is requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To insure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
3.2	DI-T-4902	First Article Inspection Test Report	____

The above DID was cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management System and Data Requirements Control List (AMSDL), must be researched to insure that only current, cleared DID's are cited on the DD Form 1423.

6.4 Technical Manuals. The requirements for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.5 First article. When a first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item should be a preproduction sample, a first article sample, a first production item, a standard production item from the contractor's current inventory (see 3.1.1) and the number of items to be tested as specified in 4.3. The contracting officer shall also include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results, and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

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6.6 Conditions for use of level B preservation. When level B preservation is specified (see 5.1), this level of protection should be reserved for the acquisition of forklift trucks for resupply worldwide under known favorable handling, transportation and storage conditions.

6.7 Subject term (key word) listing.

Electric
Fork
Lift
Truck

6.8 Changes from previous issue. Marginal notes are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army -ME
Navy -SA
Air Force -99

Preparing Activity:

Navy - SA

(Project No. 3930-0644)

Review Activities:

Army -ME
Air Force -84

User Activities:

DLA -DH

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APPENDIX

TEST METHODS

10. SCOPE

10.1 Scope. The test methods contained within this appendix are to determine whether trucks procured under this specification conform to the requirements set forth. This appendix is a mandatory part of this specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

30. TEST METHODS.

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TEST METHOD NO. 1

MAXIMUM FORK HEIGHT - FREE LIFT HEIGHT

COLLAPSED MAST HEIGHT

1. Test Course:

   (a) Level, flat surface.

2. Test Apparatus:

   (a) Tape measure.
   (b) Plumb bob.
   (c) Liquid level or clinometer.

3. Test Procedure:

   (a) Measure the true vertical distance from floor to uppermost projection of the upright assembly with mast in true vertical position as determined by a clinometer. (This is the actual collapsed mast height.)

   (b) Raise rated load until the inner mast or carriage assembly begins to exceed the specified collapsed mast height. If necessary, reposition mast to vertical position.

   (c) Measure and record the true vertical distance from the floor to the top surface of the forks at specified load center. (This is the free lift height.)

   (d) Adjust rated load to 12 inches (305 mm), plus or minus 4 inches (100 mm), above ground level and position mast in true vertical position using clinometer and plumb bob.

   (e) Attach the plumb bob to the theoretical intersection of the front surface and top surface of the forks and mark a reference point on the ground.

   (f) Elevate the test load to the maximum height and check the reference point. If necessary, reposition the mast so that the plumb bob is directly over the original reference point.

   (g) Measure and record the true vertical distance from floor to the top surface of the forks at the specified load center. (This is maximum fork height.).

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TEST METHOD NO. 2

LIFTING SPEED - LOWERING SPEED

1. Test Apparatus:

   (a) Tape measure.
   (b) Stopwatch.

2. Test Procedure:

   (a) Measure and record distance in inches from floor to top surface of forks in fully lowered position.

   (b) With battery fully charged record time in seconds required to raise rated load to maximum fork height using timing device. Record the difference between maximum fork height and the height of the forks in the lowered position.

   (c) Record time to lower rated load at maximum speed (full open valve) to 3- to 4-foot (915 to 1220 mm) height. Load to be abruptly stopped at the 3- to 4-foot (915 to 1220 mm) height.

   (d) Repeat steps (b) and (c) a minimum of 20* times, except no measurements are required.

   (e) Remove load and raise empty forks to maximum fork height.

   (f) Record time in seconds required to lower forks to lowered position.

* During quality conformance test repeat steps (b) and (c) only 3 times.

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TEST METHOD NO. 3

RIGHT ANGLE TURN

1. Test Course:

   (a) Level paved surface.
   (b) Test course in accordance with "course layout" (see Figure A-1).

2. Test Apparatus:

   (a) Tape measure.
   (b) Movable vertical barrier.
   (c) Clinometer.

3. Test Procedure:

   (a) Truck is placed with the most forward edge of centered rated load coincident to line "B". (see Figure A-1). When forks are longer than rated load, the fork tips shall touch line "B" and the fork length is included in the "A" dimension. Mast in vertical position.

   (b) Pallet is raised 6 inches (150 mm) above floor with mast true vertical measured with the clinometer.

   (c) Position movable vertical barrier within 4 inches (100 mm) of counterweight of truck.

   (d) Turn steer wheels to the extreme left position and back truck through 90-degree turn.

   (e) Measure the perpendicular distance from line "B" to the position assumed by the movable barrier. This is dimension "A".

   (f) Return truck to initial position.

   (g) Turn steer wheels to extreme right and repeat (a) through (e).

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Test Method No. 3

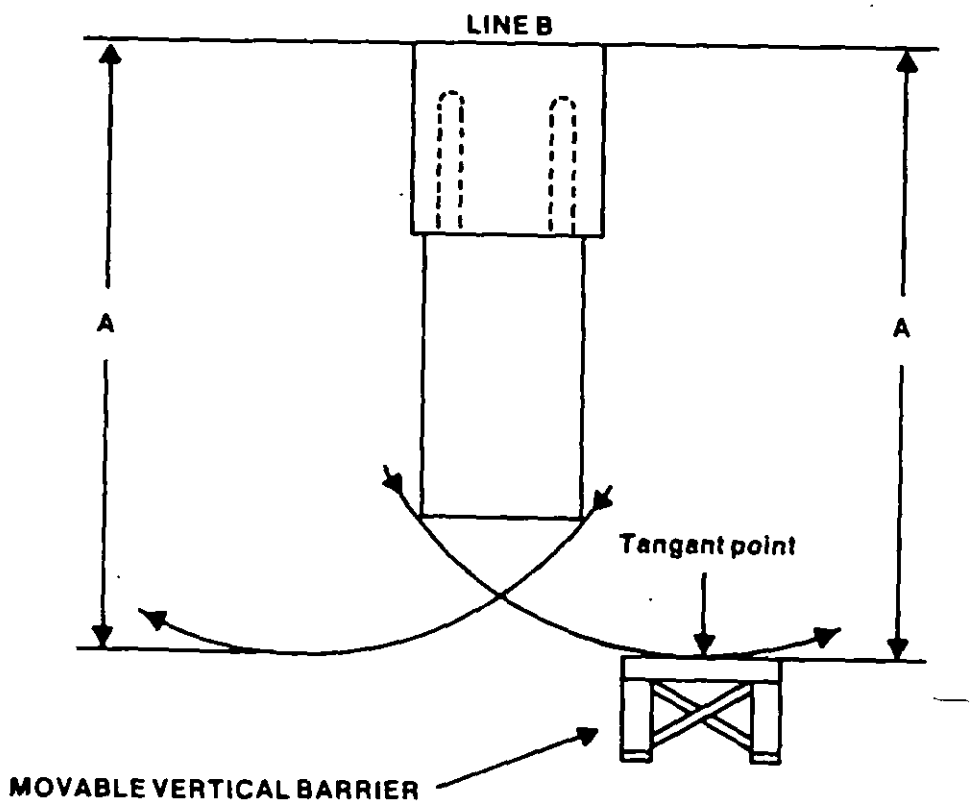


Figure A-1. COURSE LAYOUT

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TEST METHOD NO. 4

SPEED

1. Test Course:

   (a) Test course shall be a dry, level surface of sufficient length to attain maximum speed prior to entering measured distance, plus measured distance, plus sufficient stopping distance.

2. Test Apparatus:

   (a) Tape measure.
   (b) Stopwatch or electronic timer.

3. Test Procedure:

   (a) Operate the truck with fully charged battery with rated load in carry position 15 minutes prior to test.

   (b) Drive the truck a sufficient distance to attain maximum speed prior to entering the measured test course.

   (c) Record length of measured distance and time to traverse measured distance. Measured distance shall be equal to or greater than 44 feet (13 meters).

   (d) Repeat for a total of six runs, in forward gear, three in each direction, except only time measurement is required.

   (e) Repeat total of six runs in reverse gear, three in each direction, except only time measurement is required. Travel speeds in forward and reverse shall be the average of the six runs in the corresponding direction.

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TEST METHOD NO.5

SLOPE ASCENSION - PARKING BRAKE - UNDERCLEARANCE

1. Test Course:

   (a) Ramp

2. Test Apparatus:

   (a) Clinometer.
   (b) Spring scale.

3. Test Procedure:

   (a) Drive truck forward with rated load along horizontal approach and onto ramp of specified percent grade and bring to complete stop. (All wheels supported on the ramp.)

   (b) Apply parking brake and using the spring scale measure force required at end of the operating lever.

   (c) Release service brakes and observe whether truck remains stationary.

   (d) Release parking brake and start from a standstill. Proceed up ramp and onto a horizontal surface. Observe underclearance.

   (e) Repeat steps 3(a) through 3(d) without load.

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TEST METHOD NO. 6

UPRIGHT TILT

1. Test Course:

   (a) Level, flat surface.

2. Test Apparatus:

   (a) Clinometer.

3. Test Procedure:

   (a) Tilt.

      (1) Place truck with load on flat level surface, using clinometer to insure truck is level.

      (2) Raise forks to an elevation of 2 feet (610 mm) above the surface, tilt as far forward as possible, and record angle of tilt shown on clinometer located on front or rear surface of outer mast channel.

      (3) Tilt as far rearward as possible and record angle of tilt on clinometer located on front surface of outer mast channel.

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TEST METHOD NO. 7

LIFT ASSEMBLY DRIFT

1. Test Course:

   (a) Level, flat surface.

2. Test Apparatus:

   (a) Temperature thermocouples.
   (b) Clinometer.

3. Test Procedure:

   (a) Install thermocouple in the hydraulic reservoir. Heat the hydraulic oil by exercising the lift and tilt functions and maintain hydraulic oil temperature above 120°F (49°C) temperature during the duration of the test.

   (b) Raise rated load to maximum fork height obtainable with outer channel in true vertical position measured with the clinometer. Mark reference point on mast upper stage.

   (c) Record initial outer mast tilt and hydraulic oil temperatures.

   (d) Hold for specified time and record downward drift, rotational drift, and hydraulic temperature.

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TEST METHOD NO. 8

LONGITUDINAL AND LATERAL STABILITY (SEE NOTES 1 & 2)

1. Test Course:

   (a) Tilt table.

2. Test Apparatus:

   (a) Clinometer.
   (b) Plumb bob.
   (c) Tape measure.

3. Test Procedure:

   A. Longitudinal - Stacking (See Note 3).

      (a) Place truck with load on an initially level platform with axis of load wheels parallel to tilting axis of the platform. Center of mass of the test load should be on the longitudinal centerline of the truck and at the rated load center.

      (b) Elevate test load approximately 12 inches (300 mm) above platform and position mast in true vertical position using level and plumb bob, or equivalent.

      (c) Attach plumb bob to the theoretical intersection of the front surface (load face) and top surfaces of the forks and mark a reference point on platform.

      (d) Elevate the test load to the maximum height and check the reference point. If there has been any change, re-position mast so that plumb bob is directly over the original reference point.

      (e) Tilt platform to the slope specified on the individual specification sheet in the direction of the load overhang. Truck position on the platform shall be maintained by parking brakes or external locking of service brakes or other similar means. Wheel chocks shall not be used, except as specifically permitted by applicable equipment specification.

NOTE 1. During test there shall be no driver on the truck.

NOTE 2. Surface of platform may be covered with a friction increasing material such as liquid paint or adhesive applied tape containing friction material.

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B. Longitudinal Traveling (See Note 4).

(a) Place truck carrying the specified rated test standard load on an initially level platform with the axis of the load wheels parallel to the tilting axis of the platform. Center of mass of test load shall be on the longitudinal centerline of the truck and at the rated load center.

(b) Tilt mast to full rearward tilt and elevate the test load until the point of intersection of the two load carrying surfaces of the forks is 12 inches (300 mm) from the floor.

(c) Tilt the platform to the slope specified on the individual specification sheet about an axis parallel to the axis of the load wheels, and in a direction to increase the load overhang. The truck position on the platform shall be maintained by parking brake or external locking of service brakes or other similar means. Wheel chocks shall not be used, except as specifically permitted by applicable equipment specification.

C. Lateral Stacking (See Note 5).

(a) Place specified rated test load on truck in normal carrying position with center of mass of load on the longitudinal centerline of truck and at the rated load center.

(b) Position truck on initially level platform so that the tilting axis of platform is parallel to a line drawn from the center of one drive wheel tire (or outermost tire where multiple tires are used) to a point on the platform directly beneath the center of the steer wheel axis.

(c) Position the steer wheel nearest the tilting axis so that it is parallel to the tilting axis. If truck is equipped with side shifting mechanism, side shift the load to maximum position towards the platform tilting axis.

(d) Raise the load to maximum height and full rearward tilt.

(e) Tilt platform to the slope specified on the individual specification sheet. Truck position may be maintained by parking brake, external locking of service brakes or similar means, but not by means of wheel chocks, except as specifically permitted by the equipment specification.

NOTE 3. Where attachments are supplied as original equipment, this same stability test shall apply, except that the plumb line shall be attached to the underside of the carriage, attachment or load- whichever is lowest. The truck shall be equipped with the attachment, and the test shall be representative of the rated load for the combined truck and the attachment.

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D. Lateral - Traveling (See Note 6).

(a) Place truck without rated load on initially level platform. Position truck on platform so that the tilting axis of the platform is parallel to a line drawn from the center on one load wheel tire (or outermost tire where multiple tires are used) to a point on the platform directly beneath the center of the steer wheel axis.

(b) Position the steer wheel nearest the tilting axis so that it is parallel to the tilting axis.

(c) Position mast to full rearward tilt and elevate forks so that the top load carrying surface of the forks is approximately 12 inches from platform.

(d) Tilt platform to the slope specified on the individual specification sheet. Truck position on platform shall be maintained by parking brake, external locking of service brakes or similar means, but not by means of wheel chocks, except as specifically permitted by the equipment specification.

NOTE 4. Where attachments are required as original equipment, this same stability test shall apply, except the 12-inch (300 mm) measurement shall be to the undersurface of the attachment, carriage, or load - whichever is lowest. The truck shall be equipped with the attachment and the test load shall be representative of the rated load for the combined truck and attachment.

NOTE 5. Where attachments are required as original equipment this same stability test shall apply. The truck shall be equipped with the attachment and the test load shall be representative of the rated load for the combined truck and attachment.

NOTE 6. When attachments are supplied as original equipment, this same stability test shall apply, except the 12 inch (300 mm) measurement shall be made to the underside of the attachment or carriage - whichever is lowest.

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TEST METHOD NO. 9

STEERING

Part I. Wear of Steering Mechanism.*

* This part of test method to be followed before and after reliability test.

1. Test Course: Smooth, clean, dry, level surface.

2. Test Apparatus:

   (a) Piece of stiff fiber board.
   (b) Ruler.

3. Test Procedure:

   (a) Place unloaded truck on a dry, level concrete surface with its front wheels parallel to the longitudinal axis of the truck. With power steering, pump motor is to be running.

   (b) Attach to the steering column, immediately below the steering wheel and parallel to it, a piece of stiff fiber board or similar material which is semi-circular in shape and which has a radius at least one inch greater than the radius of the steering wheel.

   (c) Mark a reference point on the upper portion of the outer edge of the steering wheel; mark a corresponding point on the fiber board templet.

   (d) Turn the steering wheel counterclockwise until the wheels of the truck begin to turn. Mark a point on the fiber board templet corresponding to the reference point on the outer edge of the steering wheel.

   (e) Repeat operation (d) in the clockwise direction.

   (f) Remove the fiber board templet from the truck and place it on a flat surface. Measure the distance between the 2 points made in accordance with operations (d) and (e) above. This distance, in inches (millimeters), is the free play of the steering wheel.

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TEST METHOD NO. 9 (CONTINUED)

Part II. Steering Wheel Turning and Tangential Force Tests.

1. Test Course: Smooth, clean, dry, level surface.

2. Test Apparatus:

   (a) Steering wheel adapter pulley.
   (b) Tension dynamometer.

3. Test Procedure:

   (a) Place unloaded truck on clean, dry, level, smooth concrete with steer wheels in a straight ahead position.

   (b) Attach an adapter pulley, having a root diameter equal to the steering wheel diameter, to the spokes of the hand steering wheel.

   (c) Apply a steady force to a dynamometer attached to the adapter pulley in a counterclockwise direction and in the wheel plane until the wheel turns.

   (d) Record the maximum gage reading in 60-degree increments of steering wheel to within 15 degrees of the roadwheel steer angle.

   (e) Perform (c) and (d) in a clockwise direction.

   (f) Steps (b) through (e) may be performed with a torque reading instrument in lieu of an adapter pulley.

   (g) Turn steering wheel clockwise until steer wheels hit the stop. Apply an unbalanced tangential force of 150 lb (667 N) for non-power steering and 25 lb (111 N) for power steering to the steering wheel and maintain force for a minimum of 15 seconds. Perform this operation two more times.

   (h) Repeat step (g) in counterclockwise direction.

   (i) Measure the number of turns of the hand steering wheel required to turn the steer wheels from the extreme left position to the extreme right position. Repeat operation in the opposite direction.

   (j) Measure and record steering wheel diameter and steering wheel to seat clearance.

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TEST METHOD NO. 10

OVERHEAD SAFETY GUARD

1. Test Apparatus:

(a) Tape measure.
(b) 100 pound (45 kg) solid hardwood core (or equivalent) test weight.
(c) 500 (227 kg) pound high density cube (when a high density overhead safety guard is specified).

2. Test Procedure:

(a) Place truck on level surface.

(b) Examine safety guard and take measurements from a suitable reference point to various points on the bars or tubings in order to determine the amount of deflection at conclusion of test.

(c) Place or hold 100 pound (45 kg) test weight 5 feet (1.5 m) above the safety guard so that it will drop vertically on the safety guard between the supporting side rails of the guard. The test weight shall be positioned so that it will drop flat on the bars or tubes and not edgewise.

(d) Release test weight.

(e) Repeat steps (c) and (d) selecting points at random on safety guard upon which test weight is to be dropped a total of 10 times.

(f) At conclusion of drops, examine the overhead safety guard for fractures, permanent deformation in excess of allowable deflection.

(g) Apply static test load on the applicable specification sheet to overhead guard and observe whether guard has sufficient strength to withstand applied load(s).

The following tests are required when a high strength overhead guard is specified.

(h) See (a) & (b) above.

(i) Place or hold 500 pound (227 kg) test weight 8 feet (2.4 m) above the safety guard so that it will drop vertically on the safety guard between the supporting side rails of the guard. The test weight shall be positioned so that it will drop flat on the bars or tubes and not edgewise.

(k) See (d), (e) and (f) above.

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TEST METHOD 11

LIFTING AND TIEDOWN ATTACHMENTS

A. Lifting Attachments

1. Test Procedure:

(a) Lift the truck with battery and hold in suspension in normal transport position using slings that converge not more than 24 feet (7.3 m) above the lowest extremity of the truck. Measure clearance between each sling and the truck. Measure eye openings and clearance dimensions of the attachment. Measure pull on each attachment.

(b) Restrain the truck by anchoring the main frame and subject each attachment to a pull of 2.5 times the load it carried when initially suspended. This pull shall be applied in the direction as determined in (a) above.

(c) An alternative method of test is to lift the truck with battery as in (a) above and add weights to the main frame until the force is 2.5 times the initial force on each attachment. Hold each load for a minimum of 90 seconds.

B. Tiedown Attachments

2. Test Procedure:

(a) Inspect that tiedown attachments are so located that tiedown legs will fall within a 45-degree working cone.

(b) With the truck anchored by means other than the tiedown attachments, subject each attachment to the load specified below.

(c) Each tiedown eye shall withstand without yielding, its proportion of the following loads: 4.0 times the maximum weight of truck and battery in the direction of the longitudinal axis of the equipment, 2.25 times in the direction of the vertical axis, and 1.5 times in the direction of the lateral axis. The loads shall be applied in each direction of the axis and shall be distributed proportionally only among the tiedown eyes which would effectively resist motion if the combined directional load was applied through the center of gravity of the equipment. The test loads shall be applied for not less than 3.0 seconds and not more than 6.0 seconds.

(d) Breakage or deformation of the tiedown such that it can no longer withstand the required loads or failure to conform to 3.3.5.4 shall constitute failure of the test. Local deformation in the tiedown and nearby vicinity does not constitute failure of the test.

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TEST METHOD NO. 12

MAST ASSEMBLY OVERLOAD

1. Test Course:

   (a) Level, flat surface.
   (b) Anchor.

2. Test Apparatus:

   (a) Magnetic particle or dye penetrant capability.

3. Test Procedure:

   (a) Place truck on level surface and support it with blocks under the axle or frame to relieve strain on tires. Position mast in true vertical position, with rated load on forks in normal carrying position. External means may be utilized to hydraulically lock the tilt cylinder.

   (b) Secure truck to floor in a manner to compensate for the additional test loads to be placed on forks. Counterweight may be added to assist the holding force. Apply counterbalancing forces in a vertical direction.

   (c) With crane or another truck, place a test load of the rated capacity in position over the first test load such that the center of mass of the test load is on the truck's longitudinal centerline and at the rated load center.

   (d) Check strain or holddown devices and hydraulic system to determine whether the additional test load can be placed on forks. Tighten holddown devices if necessary.

   (e) With the crane or another truck, place the third test load carefully in position over the first two loads so that the center of mass of load is on the longitudinal centerline of the truck and at the rated load center.

   (f) When placing the overload on trucks, personnel should stand clear and crane operator should stand by to remove load if there is any evidence of immediate collapse or failure in truck structure.

   (g) The 300 percent rated load may be applied as one load in lieu of three loads as described in (c) through (f).

   (h) If truck accepts the additional test load without immediate failure, let overload stand on truck for 10 minutes.

   (i) Remove loads.

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(j) Inspect truck structure, frame, mast assembly, the hydraulic system for evidence of deformation, fractures, leakage in hydraulic system, broken welds, etc.

(k) Inspect forks a minimum of 6 inches (150 mm) each side of heel by magnetic particle or dye penetrant method.

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TEST METHOD NO. 13

STOPPING DISTANCE

1. Test Course:

   (a) Test course shall be a dry, level surface of sufficient length to obtain maximum speed plus sufficient stopping distance.

2. Test Apparatus:

   (a) Tape measure.

3. Test Procedure:

   (a) Operate truck for 15 minutes prior to test.

   (b) Run the truck carrying rated load in rearward direction. Load shall be in normal carrying position.

   (c) After reaching top speed, the brakes shall be applied.

   (d) Measure the distance traveled by the truck after application of the brake to the final stopping point. Record this distance.

   (e) Perform operations (c) and (d) a second and third time.

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TEST METHOD NO. 14

SERVICE BRAKE FORCE AND TOWING HOOK STRENGTH

1. Test Course:

Level, clean, dry, concrete, having a coefficient of friction not less than .50.

2. Test Apparatus:

(a) Pedal force gage.

(b) Tension dynamometer

(c) Block and tackle

3. Test Procedure:

A. Brake Pressure Test.

(a) Attach the force gage to the brake pedal in a manner which enables pressure to be applied to the face of the gage.

(b) Apply 250 pound (1110 N) forcetom the brake pedal, hold for one minute and release. Repeat this operation 10 times.

(c) Inspect for component failure.

B. Towing Hook Strength.

(a) Determine gross weight of truck with rated load.

(b) Attach force gage to brake pedal in a manner which will enable force to be applied to the face of the gage.

(c) Attach a tension dynamometer to the rear towing eye.

(d) Attach one end of a block and tackle, or similar mechanical device, to the tension dynamometer, and the other end to an anchoring device.

(e) With transmission in neutral, tow truck with block and tackle or similar mechanical device.

(f) Apply sufficient brake pedal force to produce a force recorded on dynamometer equal to 30%* of the total weight of truck plus rated load and record brake pedal and dynamometer readings.

(g) Examine towing eye for evidence of permanent deformation.

* 50% if 4 wheel brakes.

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TEST METHOD NO. 15

MAINTAINABILITY

1. Test Course:

Shop area of sufficient size to permit unobstructed surveillance of maintenance.

2. Test Apparatus:

(a) Common handtools.
(b) Special tools (if furnished with truck).

3. Test Procedure:

(a) Examine truck for accessibility of major assemblies.

(b) Examine position of drains with respect to accessibility.

(c) Using the tools, demonstrate that major assemblies are accessible for repair and maintenance.

(d) Demonstrate both the accessibility of drains and the path of discharge of lubricants by activating drainage controls.

(e) Perform and time each operation on the maintenance operations list (see 3.1.3.7.1).

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TEST METHOD NO. 16

LOAD BACK REST TEST

1. Test Course:

   (a) Shop area.

2. Test Apparatus:

   (a) Test weight.
   (b) Stop watch.

3. Test Conditions: A completely assembled truck may be used for this test or it may be performed as a bench test using fork carriage and back rest. A force equal to the rated load shall be applied at right angles to the back rest and uniformly distributed over the entire area comprising the back rest.

4. Test Procedure:

   (a) Place a metal sheet over the total area comprising the back rest.

   (b) Apply a weight equal to the rated load, at right angle to the back rest area.

   (c) Hold weight on truck's load back rest for not less than three (3) minutes.

   (d) Remove load and inspect the entire back rest, for cracks, breaks, failures or any permanent deformations of any members of the back rest.

   (e) Check back rest for ease of removal after loading.

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TEST METHOD NO. 17

TIRE LOADING

1. Test Apparatus:

   (a) Platform scales.

2. Test Procedure:

   (a) Determine the weight supported by each trail wheel without a load on the forks by driving the trail wheels onto a platform scale or jacking them up with two axle scales. When the trail wheels are weighed together, the total weight should be divided by the number of trail tires. The upright shall be in a vertical position and the forks approximately 12 inches (305 mm) above the floor.

   (b) Determine the weight supported by the load wheels with rated load in a carry position on the forks by driving the load wheels on a platform scale or jacking up with two axle scales. When a side-shift mechanism is not provided and the load wheels are weighed together, divide the total weight by the number of load carrying tires. When a side-shift mechanism is provided, the load shall be shifted to the extreme left and the weight carried by each load wheel determined separately. Repeat the procedure with the load shifted to the extreme right. The upright shall be vertical and the load shall be approximately 12 inches (304 mm) above the floor.

   (c) Repeat steps (a) and (b) with rated load on the forks centered on the carriage.

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TEST METHOD NO. 18

POWER CONSUMPTION TEST

1. Test Course:

Dry, level area of sufficient size to perform test described below.

2. Test Apparatus:

(a) Stop watch.

3. Test Procedure:

(a) The truck shall be cycled continuously for 7.5 hours with one battery only (battery shall not be charged during the test).

(b) Each cycle detailed below shall take 2.75 +/- 0.25 minutes.

(c) Cycles shall be timed and recorded.

(d) One cycle consists of the following operations on the test course.

(1) Pick up capacity load "A" at area "X" and place it at area "Z".

(2) Back up to area "Y".

(3) Pick up capacity load "B" at area "Y" and place it on load "A" at area "Z".

(4) Back out and drop forks.

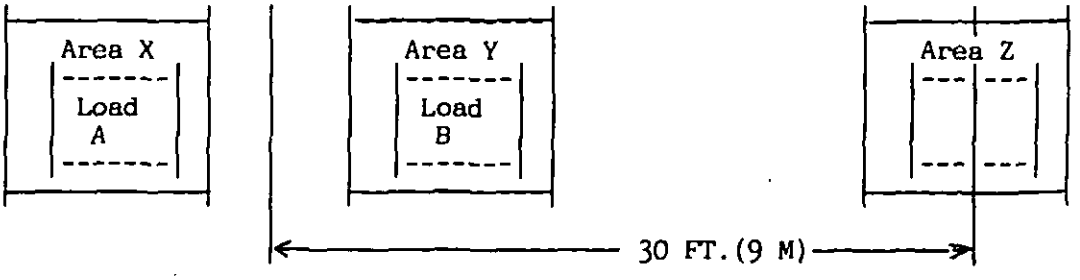
(5) Raise forks, pick up load "B" and place it at area "Y".

(6) Back out to area "Z":

(7) Pick up load and place it at area "X".

(e) Record number of cycles per hour.

(f) Observe that truck negotiated test course for 7.5 hours.



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TEST METHOD NO. 19

WEIGHING DEVICE

1. Test Area:

   (a) Level, flat surface.

2. Test Apparatus:

   (a) Test load.

3. Test Procedure:

   (a) With the truck at rest on a smooth, level surface and with the mast vertical, the unloaded forks shall be elevated to a height of 20 - 24 inches (510 - 610 mm). By use of the control valve, the forks and carriage shall be dropped 1 to 4 inches (25 to 100 mm) and abruptly stopped.

   (b) Repeat (a) four times.

   (c) Adjust tare to show zero load where readout most consistently came to rest during (b) within approximately the first ten seconds and mark the zero position.

   (d) With the truck at rest on a smooth level surface and with the mast vertical the forks shall elevate a calibrated load equal to the rated capacity of the truck plus or minus 1-percent to a height of 20 - 24 inches (510 - 610 mm) and through the control valve the load shall be dropped from 1 to 4 inches (25 to 100 mm) and then stopped abruptly.

   (e) Repeat (d) four times.

   (f) Failure of the weighing device to weigh the calibrated load during (e) within the required accuracy shall be cause for rejection.

   (g) With the truck at rest on a smooth level surface, mast vertical, forks 20 inches (510 mm) above the surface, a calibrated load equal to 150 percent of the truck rated capacity shall be placed on the forks.

   (h) Remove the load and repeat the weight accuracy test procedure prescribed in (g).

   (i) Failure of the weighing device to weigh the calibrated load within the required accuracy and repeatability shall be cause for rejection.

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TEST METHOD NO. 20

RAIN TEST

1. Test Apparatus:

   (a) Timing Device.
   (b) Measuring Device (U.S. Weather Bureau type preferred).
   (c) Spray Fixture.

2. Test Procedure:

   (a) Subject the truck to a simulated rainfall of not less than 4 inches (100 mm) per hour. The simulated rainfall shall impinge on the truck at an angle of 45 degrees plus or minus 5 degrees with a force equivalent to a 30 mph (48 km/hr) wind. The simulated rainfall shall dispense uniformly over the surface of the truck.

   (b) Expose the top and left side of the truck to the rain described above for a period of 5 minutes.

   (c) Repeat (b) with top and front exposed.

   (d) Repeat (b) with top and right side exposed.

   (e) Repeat (b) with top and rear exposed.

   (f) Examine all instruments and electric controls for evidence of moisture.

   (g) Demonstrate operation of the truck.

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TEST METHOD NO. 21

FRAME RACKING

1. Test Description: Test is described under test procedure. This test may be performed before or after truck is completely assembled. When performed as a bench test prior to assembly, a truck of similar rating and characteristics may be used to determine the maximum static forces to be applied to support area.

2. Test Procedure:

(a) Determine by weighing or other means, the vertical static force exerted by the completely assembled truck at the support areas of the frame when carrying rated load and without load.

(b) Secure all but one support area of the frame of truck in a jig or similar device. The frame shall be in a horizontal position.

(c) Apply a vertical force to the unsecured support area equal to 200 percent of the maximum static force obtained in step (a) for that support area.

(d) Repeat steps (b) and (c) for the other support areas.

(e) Examine truck frame for evidence of fractures, permanent deformation or distortions.

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TEST METHOD NO. 22

TEMPERATURE STABILIZATION

1. Test Course:

   (a) Shall be in accordance with Figure 1.

2. Test Apparatus:

   (a) Thermocouple(s).
   (b) Pyrometer.
   (c) Oven.
   (d) Thermometer

3. Test Procedure:

   (a) Remove thermal switches and place in oven. Slowly increase temperature until switches open. Read and record temperatures at which thermal switches open.

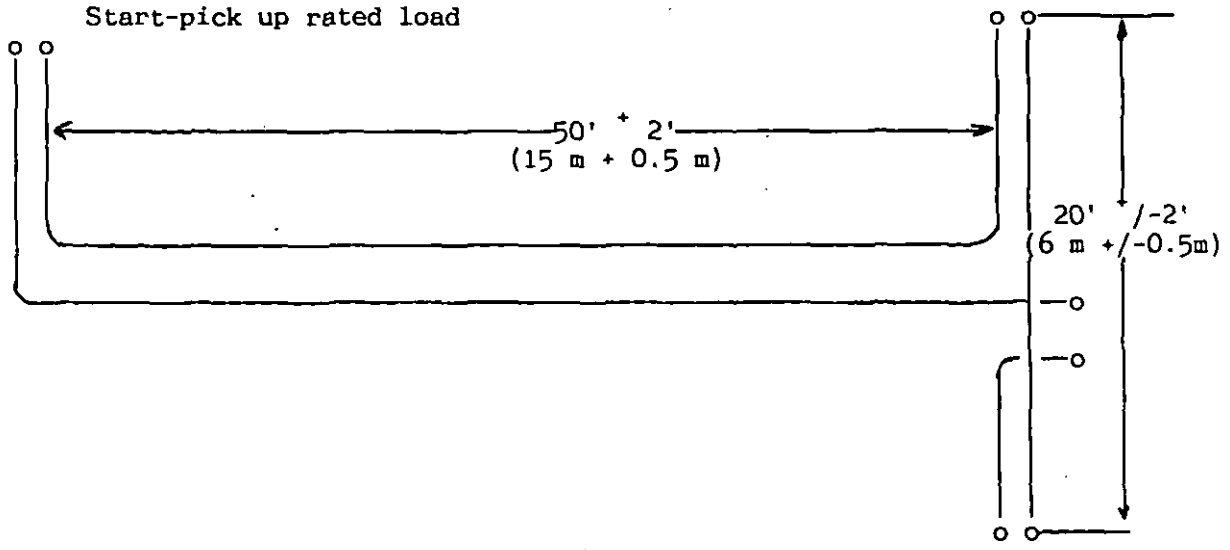
   (b) Reinstall thermal switches on truck and install thermocouple(s) adjacent to thermal switches. Additional thermocouple shall be submerged in the hydraulic reservoir.

   (c) When required, place truck in facility where ambient temperature can be maintained.

   (d) Operate truck continuously at low speed over the test course. Read and record temperatures including ambient at intervals not more than 15 minutes. Continue operation on test course until all critical temperatures stabilize for two consecutive readings.

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FIGURE 1 (TEST METHOD NO. 22)



Reverse, Low Speed, W/Rated Load
Forward, Low Speed, W/Rated Load
Reverse, Low Speed, W/O Rated Load
Forward, Low Speed, W/O Rated Load

Note: On alternate laps, the load should be picked up where it was previously deposited and redeposited where it was previously picked up. The load shall be raised to full lift height before it is deposited.

TEST COURSE FOR ELECTRIC (SPARK ENCLOSED) FORK LIFT TRUCK

(MOTOR AND RESISTOR HOUSING HEAT STABILIZATION)

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TEST METHOD NO.23

HYDRAULIC FILTER

1. Test Apparatus:

(a) Pressure gages.
(b) Temperature gages.
(c) Flowmeter.
(d) Timer.
(e) Oil conforming to MIL-L-2104, Grade 10.

2. Test Procedure:

(a) Filter housing proof test.

(1) Block housing outlet port. Pressurize the filter housing to 100 psi (69 KPa) (30 psi (20 KPa) for suction filters) minimum, and hold for not less than 60 seconds. Examine for leaks. This test is not required when hydraulic reservoir serves as filter housing.

(b) Pressure drop.

(1) Install the filter with a new element in the test stand or on the truck. Establish maximum system flow through the filter at an oil temperature of 150°F (65°C), plus or minus 5°F (3°C). Measure and record the pressure drop across the filter assembly.

(c) Bypass valve pressure drop.

(1) Plug the filter element chamber oil exit or otherwise obstruct all flow through the filter element. Reinstall the housing in the test circuit with outlet port open. Establish and maintain the oil temperature at 150°F (65°C). Increase upstream pressure to 10 psi (69 KPa) more than that determined in (b) above (1.75 psi (12 KPa) for suction filters). Observe contamination indicator. Increase upstream pressure until the indicator enters bypass area or bypass valve leakage exceeds 5 percent of maximum system flow, whichever occurs first. Observe and record pressure. Increase pressure until bypass is fully open. Establish maximum system flow. Observe and record pressure and flow.

(2) Use any combination of on-the-truck and off-the-truck filtration for cleanup.

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(3) When bottle sampling is used, extract a sample for analysis from a turbulent point in the system with the pump running or within 2 minutes of shutdown. When dynamic sampling is used, connect the counter sensor to the system, using a bypass line and a branch line. The bypass line shall connect to the system, have a flow rate not less than 5 times the sensor flow rate, and may return fluid directly to the reservoir. The branch line to the sensor shall tee into the bypass line and shall be not more than 12 inches (305 mm) in length. Any flow restriction devices shall be placed down-stream of the particle counter sensor. Particle counts should be allowed to stabilize before recording counts.

(e) Continue cleanup procedures as described in respective paragraphs above until the average of not less than three consecutive counts is less than 2000 particles per milliliter larger than 10 micrometers and less than 20 particles per milliliter larger than 20 micrometers.

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TEST METHOD NO. 24

CYLINDER PROOF

1. Test Apparatus:

   (a) Hydraulic test bench or truck capable of developing two times system working pressure.
   (b) Pressure gage.
   (c) Thermocouple.

2. Test Procedure:

   (a) Double acting cylinder.

      (1) Position and mechanically hold the piston at the approximate midpoint of the cylinder.

      (2) Fill both sides of the piston with oil at a minimum of 120°F (49°C).

      (3) With the head end port capped, apply an oil pressure equal to 1.5 times the working pressure to the rod end of the piston for minimum of 60 seconds. Release pressure.

      (4) With the rod end port capped, apply an oil pressure equal to 1.5 times the working pressure to the head end of the piston for minimum of 60 seconds. Release pressure.

   (b) Single acting cylinder.

      (1) Extend cylinder to maximum extension by filling with oil at a minimum of 120°F(49°C).

      (2) Apply oil pressure equal to 1.5 times the working pressure at the inlet port for a minimum of 60 seconds. Release pressure.

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TEST METHOD NO. 25

HYDRAULIC SYSTEM CLEANLINESS

1. Test Apparatus:

   (a) Liquid Automatic Particle Counter.

2. Test Procedure:

   (a) When hydraulic line volume exceeds 1/2 the volume of the related cylinder, connect the hydraulic lines on the cylinders to bypass related components.

   (b) Run the pump pumping the oil through each circuit for not less than 5 minutes. Perform the particle count as described in (d) below.

   (c) Connect each line to its related component. Operate each previously bypassed function of the system for 5 minutes continuously through full stroke and at maximum rates. Perform the particle count as described in (d) below.

   (d) The procedure for determining contamination shall be as follows:

      (1) Contamination levels shall be determined by particle counts. The particle counter used for evaluation of the samples shall be calibrated in accordance with ANSI Recommended Standard Method for Calibration of Liquid Automatic Particle Counter using "AC" Fine Test Dust (ANSI B93.28). The required counts shall be the average of not less than three consecutive counts. Samples may be taken from the reservoir or upstream of the filter(s) used for cleanup.

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TEST METHOD NO. 26

HYDRAULIC FITTINGS

1. Test Apparatus:

   (a) Common handtools.

2. Test Procedure:

   (a) Remove one hydraulic hose assembly which has permanent type fittings.

   (b) Replace the hose assembly with an equivalent field attachable fitting only to assure interchangeability.

   (c) Repeat (a) and (b) above until each hose assembly with permanent type fittings is replaced with a hose assembly with reuseable fittings.

   (d) After every hose assembly with permanent fittings has been replaced with a hose assembly with reuseable fittings, reinstall the original hose assemblies on the truck.

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TEST METHOD NO. 27

RELIABILITY TEST

1. Test Course:

(a) Layout of course. The test course shall be set up equivalent to the provisions of Figure A-2. Aisle widths shall not exceed the dimensions shown in Table I. The course perimeter shall be demarcated with suitable barriers or indicators. Rubber pylons or other suitable marking shall be placed on all corners and other critical points to assist in keeping the truck being tested within the course limits.

TABLE I

AISLE WIDTHS FOR TEST COURSE (FT/M)	
Capacity (lb/kg)	Solid Tired Trucks
2000/908	10/3
4000/1816	12/3.6
6000/2724	14/4.3
8000/3632	14/4.3
10000/4540	14/4.3

(b) Obstacle inclusion. The obstacle test shall be set up in accordance with the provisions of Figure A-3, as a part of the test course. Iron, wood, cement or steel blocks may be used. A guideline 15 feet (4.6 m) long shall be provided to the left of the centerline of the course and parallel to it. The inner edge of this guideline shall be at a distance from the course centerline equal to one-half of the truck width or one-half of the load width (whichever is greater) plus 6 inches (150 mm). This spacing and arrangement will allow the left wheels of the truck under test to pass over the first block and then the right wheels to pass over the second block while the truck is moving forward in a straight line parallel to the guideline.

(c) Ramp inclusion. The course shall include a ramp constructed in accordance with the provisions of Figure A-4.

(d) Course surface. The course shall be paved with concrete having a fine float finish, or equivalent, having a road resistance of approximately 30-50 pounds per ton (16-26 kg per metric ton). It shall be dry, clean and free of any nonplanned obstacles or foreign material while conducting the reliability test.

(e) Pallet areas. Pallet loads in the first pallet area shall be stacked in accordance with Figure A-5. They shall be placed 2 inches (50mm) apart in a direction parallel to, and touching in a direction perpendicular to the axis of the test course at that point.

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Side pallet stacks shall be high enough that the bottom of the top face of the top pallet in each stack is 6 inches (150 mm) below the maximum lift height of the vehicle under test. The stack marked "X" on Figure A-2 shall be high enough that the bottom of the top face of the pallet, when placed upon it, shall be 6 inches (150 mm) below the maximum lift height of the vehicle under test.

Pallet loads in the second pallet area shall be stacked in accordance with Figure A-6. They shall be placed 2 inches (50 mm) apart in a direction parallel to, and touching in a direction perpendicular to the axis of the test course at that point. Side pallet stacks shall be high enough so that the bottom of the top pallet will be approximately as high as the maximum lift height of the truck under test. The stack marked "Y" on Figure A-2 shall be high enough that the bottom of the top face of the pallet, when placed upon it, shall be at the midpoint between ground level and the maximum lift height of the vehicle under test.

Pallet loads in the third pallet area shall be stacked in accordance with Figure A-7. They shall be placed 2 inches (50 mm) apart, positioned as shown on Figure A-2,in a direction parallel to, and touching in a direction perpendicular to the axis of the test course at that point. Side pallet stacks shall be one pallet load (48 inches (1220 mm) plus the height of the pallet) high.

2. Test Procedure:

Each truck shall complete 2,000 circuits without failure of the test course as detailed below.

(a) The light switch shall be turned "off" and "on" at the beginning of each lap and upon entering each pallet area.

(b) The truck under test shall begin each lap at the point labeled "START" on Figure A-2. As the operator starts the truck, he shall operate the horn for approximately 1 second. On lap number 1, the truck shall proceed in a forward direction, following the centerline along the portion of the course marked "A" until it reaches the first pallet area. The rated load will have been placed in this location on the stack marked "X".

(c) The truck shall make a 90-degree right-hand turn in one motion without backing, proceed to the face of the pallet stack marked "X", raise forks to maximum fork height so hydraulic relief valve is activated, then adjust fork height and remove the loaded pallet. The truck shall back out of the stack aisle into the main aisle making a 90-degree left-hand turn in one motion, without going forward, and proceed rearward along the main aisle in the portion of the course marked "B" until it reaches the second pallet area.

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(d) The truck shall proceed rearward to a point beyond the pallet area opening, reverse its direction, and move forward to the face of the pallet stack marked "Y"; making the 90-degree right-hand turn in one motion without backing. The rated load shall be placed on top of the pallet stack marked "Y".

(e) The truck shall back out of the stack aisle into the main aisle, making the 90-degree left-hand turn in one motion without going forward, and proceed forward along the main aisle in the portion of the course marked "C" until it reaches the third pallet area.

(f) The truck shall make a 90-degree right-hand turn in one motion without backing and pick up rated load which has been placed in the location marked "Z". The truck shall back out into the main aisle, making the 90-degree right-hand turn in one motion, and proceed further along the main aisle in the portion of the course marked "D" until it reaches the portion of the course marked "E", containing the obstacle blocks.

(g) The obstacle blocks shall be traversed by the truck being driven through this area in a straight line so that its left wheels pass over the first block and the right wheels pass over the second block. This part of the test may be conducted at low speed with the transmission in low range. However, the truck shall not be stopped prior to or while going over the blocks.

(h) The truck shall proceed further along the main aisle in the portions of the course marked "F" and "G" until it reaches the position on the ramp labeled "Stopping Point". The driver shall stop the truck and then resume ascending the ramp. After the top of the ramp is reached, proceed along the main aisle in the portion of the course marked "J", "K", and "L".

(i) When the truck reaches the portion of the course marked "M", the operator shall stop. On every other lap, the operator shall turn off the master switch, then turn "ON". After completion of this portion to the test,proceed along the main aisle and start the next cycle. All stops shall be at the maximum safe deceleration rate. The operator shall actuate and release the parking brake at least once during each circuit.

(j) The truck shall then reenter the portion of the course marked "A". In portions "A", "B", "C", and "D", the procedure shall be the same as described in preceding paragraphs, except that the truck shall place a test load at each point where it was previously indicated that the truck picked one up, and pick up a load where it was previously indicated one was placed. Maneuvers in the balance of the course beyond the portion marked "D" shall be the same as described in preceding paragraphs.

(k) One-half of the total number of laps shall be traversed in the opposite direction with all operations reversed accordingly. This may be accomplished after completion of each 8-hour operating day or

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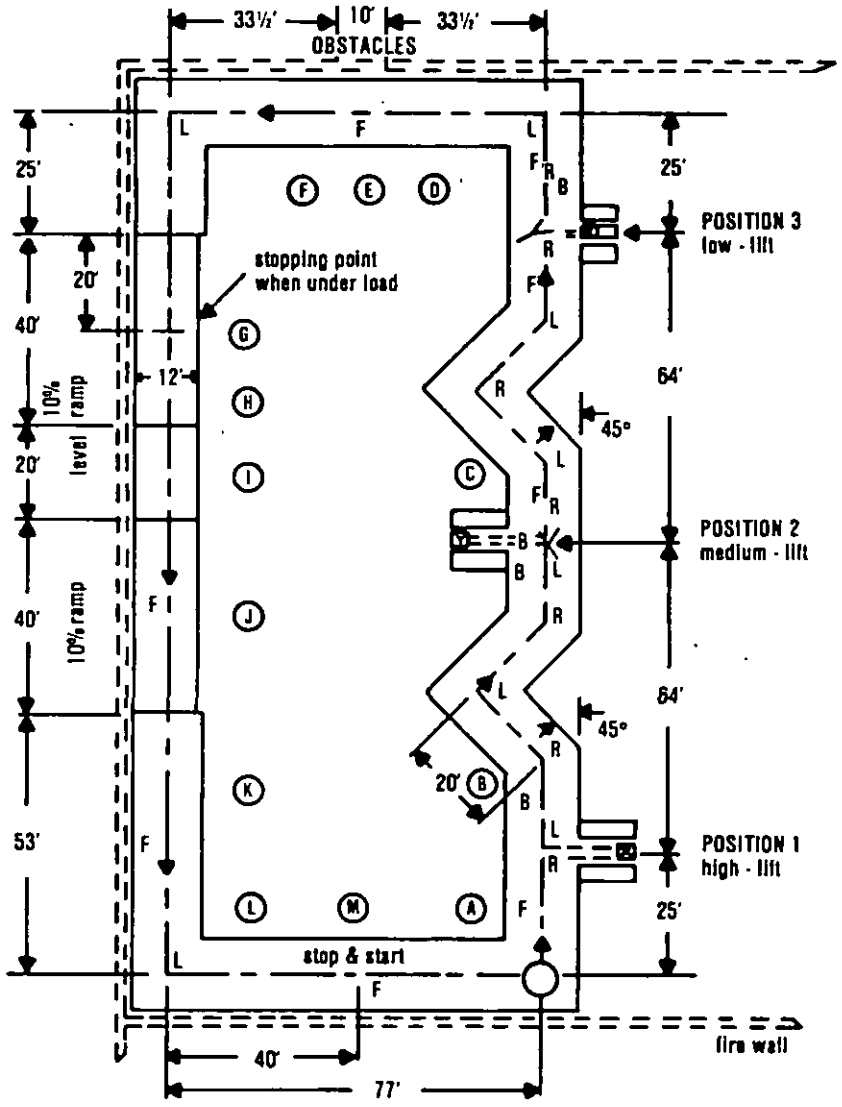
multiple thereof. That is, 50 percent of the laps shall be accomplished in the clockwise direction and 50 percent in the counterclockwise direction or vice versa. The test course need not be rearranged for the reverse travel.

(l) "The Hourly Time Record Sheet" (page 101) shall be filled out for each operating hour of the performance test. Average lap speed per hour shall be recorded.

(m) Maintenance and inspections shall be performed in accordance with the maintenance service and inspection sheet (page 98).

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TYPICAL INDOOR TEST COURSE

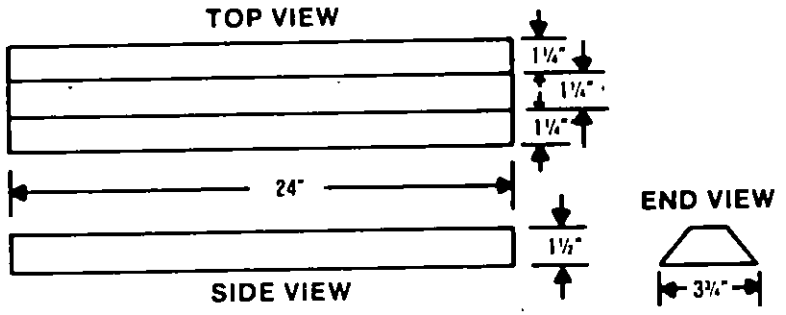


LEGEND:
F = Forward
B = Backward
L = Left turn (facing forward)
R = Right turn (facing forward)
(A) ETC. = Section designation

FT	M
10	3
12	3.6
20	6
25	7.5
33.5	10
40	12
53	16
64	19
77	23

FIGURE A-2

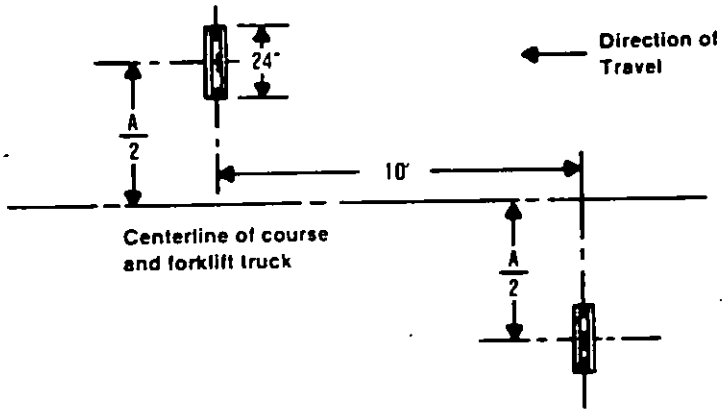
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IN	MM
1.25	32
1.5	38
3.75	95
24	609
120	3048

2 Blocks required
Blocks to be attached to test course as shown below

LAYOUT OF OBSTACLE COURSE



A = Distance between centerline of driving wheels

Figure A-3. DETAILS OF OBSTACLE CONSTRUCTION FOR TRUCK, FORK, LIFT, SOLID RUBBER TIRES

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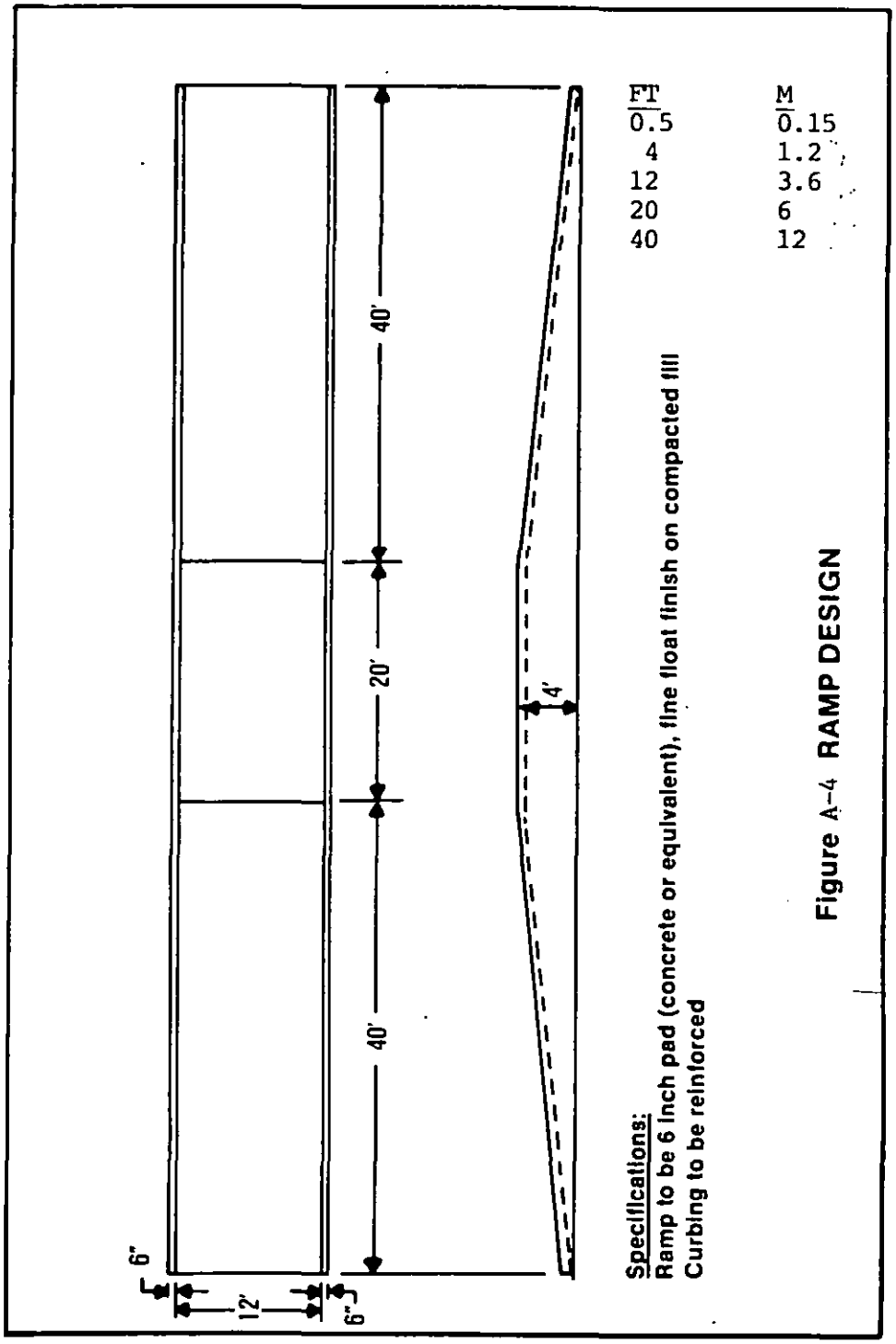
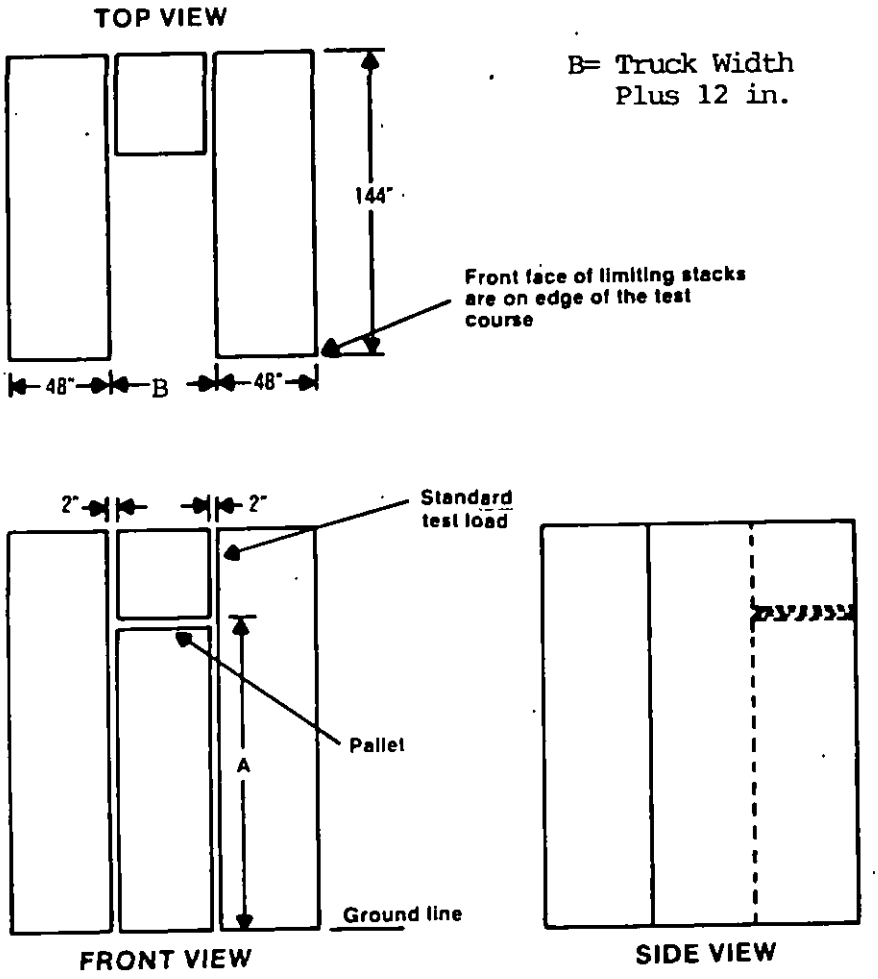


Figure A-4 RAMP DESIGN

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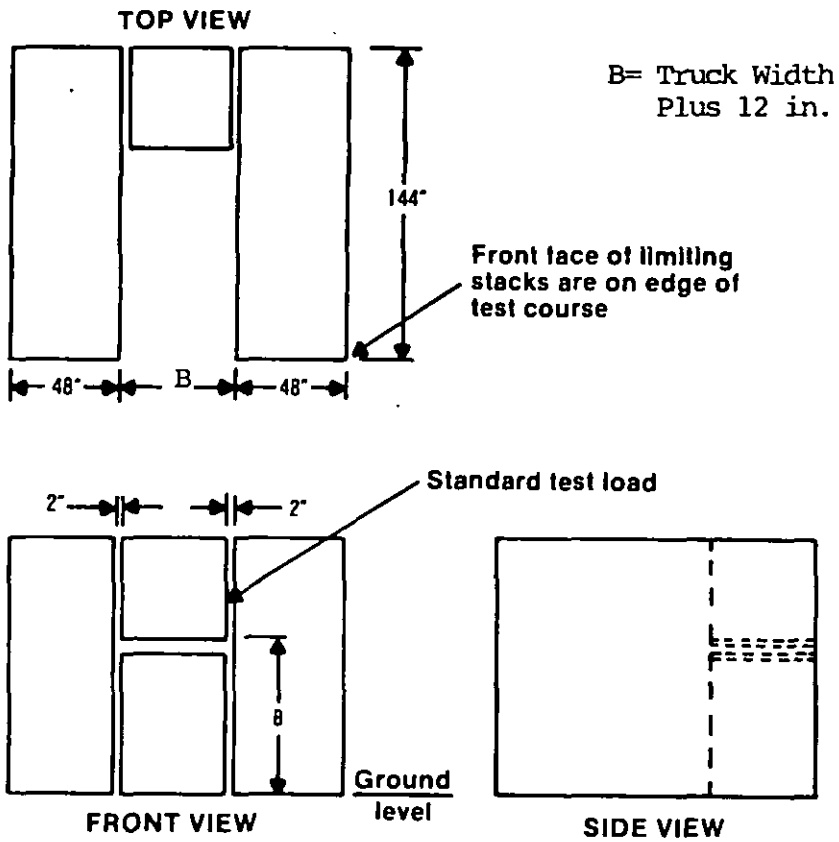
IN	MM
2	51
12	284
48	1219
144	3660

A = Maximum lift height for forks less 6 inches

Support for standard load should be of suitable strength to sustain test load. Limiting stacks may be pallets, pallet racks or both.

Figure A-5. STACK LAYOUT - POSITION 1 SOLID RUBBER TIRED TRUCKS

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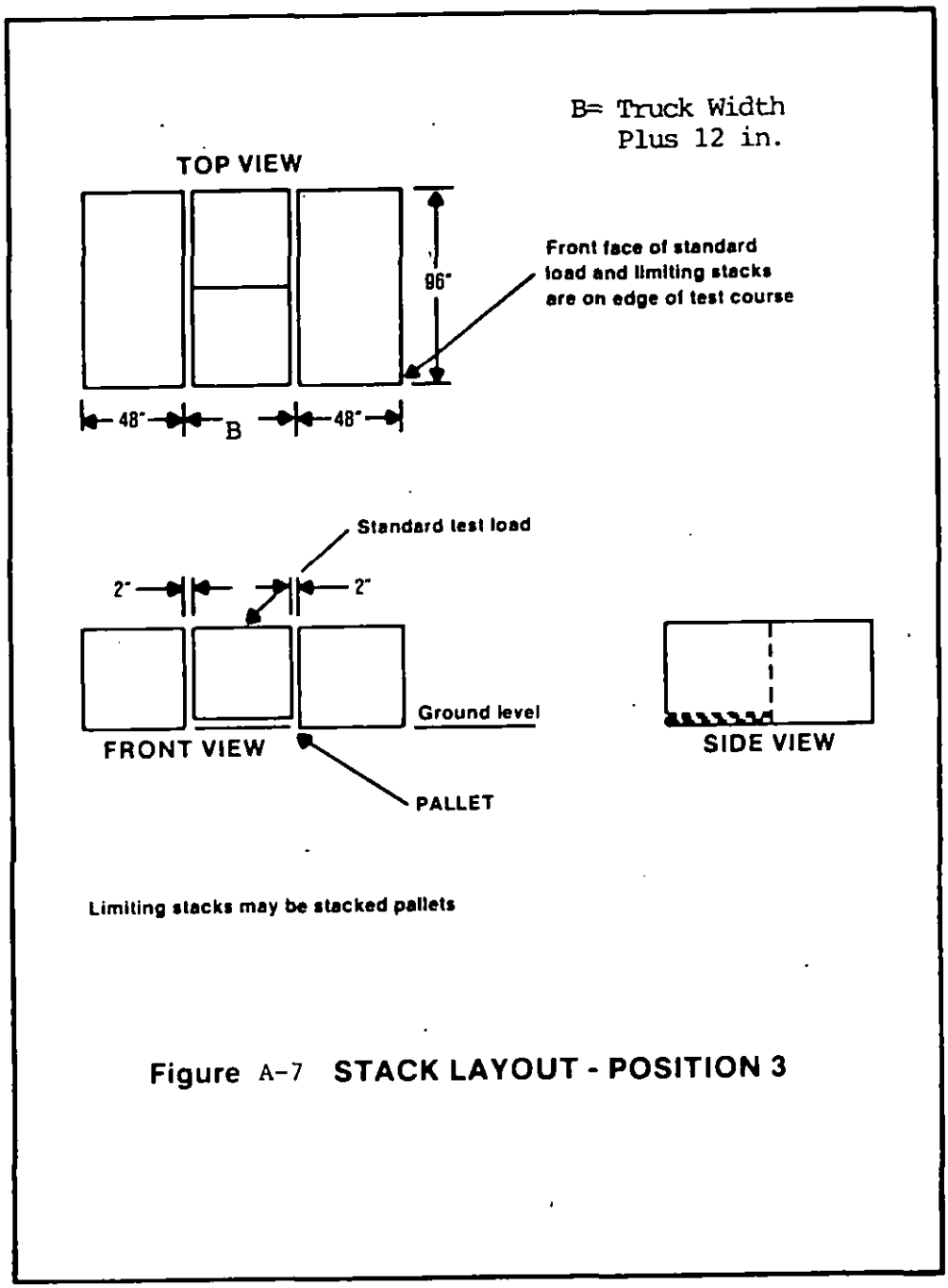
IN	MM
2	51
12	284
48	1219
144	3660

B = One-half maximum lift height of forks

Support for standard load should be of suitable strength to sustain test load. Limiting stacks may be pallets, pallet racks, or both.

Figure A-6 STACK LAYOUT - POSITION 2 SOLID RUBBER TIRED TRUCKS

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IN	MM
2	51
12	284
48	1219
96	2438

Figure A-7 STACK LAYOUT - POSITION 3

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MAINTENANCE SERVICE AND INSPECTION SCHEDULE
MATERIALS HANDLING EQUIPMENT-ELECTRIC

The following inspections and checks shall be made prior to endurance testing and prior to the post testing. Items 1 through 12 shall be accomplished (Test Method No. 22) after completion of the endurance test.

1. Check all instruments for proper operation--check free play in steering--check horn operation.
2. Check brake pedal travel--check fluid level--adjust brakes if necessary--check safety seat brake--check linkage and operation-- inflate tires to proper pressure (cold) and remove all foreign matter from tires.
3. Inspect all lights, windshield wipers and mirror. (When applicable.)
4. Check battery--water level and specific gravity.
5. Check hydraulic oil dipstick--add oil if necessary.
6. Inspect all safety equipment--fire extinguishers, overhead guards; check for cracked or bent forks.
7. Check controller contact tips, switches, operation.
8. Check motors' brush wear, spring tension, commutator wear--blow out carbon dust.
9. Check acceleration control--speed--time delay--general operation.
10. Check cylinders for leaks and wear--wipe off.
11. Inspect all wires and terminals; tighten and replace where necessary.
12. Lubricate chassis using manufacturer's reference charts--replace missing or broken fittings where necessary--make all necessary adjustments.
13. Inspect drive unit--check for noise--check oil level--fill if necessary--do not overfill.
14. Inspect steering gear--check oil level--fill if necessary--do not overfill.
15. Inspect carriage and channel assembly (rollers, carriage and mast) for wear and damage--adjust and align if necessary.
16. Inspect lift chains for wear and damage.
17. Check brake lining and drums, check wheel cylinders.
18. Wheel bearings--repair and adjust, tighten all hub flange bolts.

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19. Hydraulic system--check for noise, leaks, and operation--tighten all lines and fittings.

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HOURLY TIME RECORD SHEET

Date _____

Truck Manufacturer _____ Clock Readings: _____

Model _____ Test Location _____ Start _____

Serial No. _____ Recorded by _____ Finish _____

Time	Laps	Average Lap Time	Comments

Instructions:

1. Start decimal minute stop watch at zero at beginning of each hour.
2. Record watch reading at end of each hour and record elapsed time.

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TEST METHOD NO. 28

BATTERY PROTECTOR

1. Test Apparatus:

   (a) Potentiometer

2. Test Procedure:

   (a) Verify with a potentiometer that battery voltage is 2.12 volts times the number of cells and that the signal light or fuel gauge indicates "full charge". Operate the lift control.

   (b) Increase the value of potentiometer resistance observing voltage across the battery protector voltage terminal to 1.7 volts per cell and after one half hour observe that the warning light indicating "low state of discharge" has been activated. After a short interval of time (not to exceed 15 minutes) operate lift control and confirm lift will not operate.

   (c) Unplug and replug the battery and verify the lift control will not operate. Unplug the battery, increase the voltage with the potentiometer to 2.06 volts and replug the battery. Observe that the lift control will operate and warning lights are reset.

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TEST METHOD NO. 29

FOR COLD STORAGE CONDITIONED TRUCKS

1. Test Description:

Test is described under test procedure. Test is to verify truck will operate in refrigerated spaces and also when truck is cold and in warm ambient temperature.

2. Test Procedure:

(a) Cold Soak forklift in refrigerator for at least 24 hours. (Ambient temperature -15°F (-25°C) or colder). After cold soaking operate forklift including steering, forward and reverse motion, lift mechanism, tilt, side shift, horn and lights in cold ambient temperature.

(b) Remove forklift from refrigerator, allow truck 2 hours to warm in ambient temperature of at least 60°F (42°C). After warming operate forklift as in paragraph (a).

(c) Malfunction of any instrument, control or component or failure to complete any operation or moisture accumulation in any electrical compartment shall be cause for rejection.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-21869E	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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